Proposal Part B: technical description

SUSTAINABLE, RELIABLE AND EFFICIENT FLOATING PV POWER PLANTS (SURE)

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No. *	Participant organization name	Short name	Country
1	Institute For Energy Technology	IFE	Norway
2	Ciel et Terre International	CTI	France
3	Laketricity	LKY	France
4	ZIMMERMANN PV-Stahlbau GmbH & Co. KG	ZIM	Germany
5	BayWa r.e. Solar Projects GmbH	BayWa	Germany
6	Sunlit Sea AS	Sunlit	Norway
7	UAB Metsolar	Metsolar	Lithuania
8	Nederlandse organisatie voor toegepast natuurvwetenschappelijk onderzoek TNO	TNO	Netherlands
9	Fraunhofer Gesellschaft zur Forderung der Angewandten Forschung EV	Fraunhofer	Germany
10	Foundation Deltares	Deltares	Netherlands
11	Association Compaz	Compáz	Switzerland
12	SABIC Global Technologies BV	SABIC	Netherlands

List of abbreviations

Abbreviation	Definition	Abbreviation	Definition
aIRT	Aerial Infrared Thermography	**IV**	Current-Voltage
BoS	Balance of Systems	**KER**	Key Exploitable Results
CAPEX	Capital Expenditures	**LCA**	Life Cycle Analysis
CFD	Computational Fluid Dynamics	**LCOE**	Levelized Cost of Electricity
D#	Deliverable Number	**O&M**	Operation and Maintenance
EIA	Environmental Impact Assessment	**OPEX**	Operational expenditures
ESIA	Environmental and Social Impact Assessment	**PID**	Potential-Induced Degradation
EYA	Energy Yield Assessment	**PLR**	Performance Loss Rates
FEM	Finite-Element Modelling	**PP**	Polypropylene
FP	Funded Project	**PR**	Performance Ratio
FPV	Floating Photovoltaics	**(m)RBI**	(maximum) Ratio of Blocked Irradiation
FTE	Full-time Employees	**RE**	Renewable Energy
GLM	General Lake Model	**SBoS**	Structural Balance of Systems
GMB	Group Model Building	**SDG**	Sustainable Development Goal
GPV	Ground-Mounted Photovoltaics	**SO**	Specific Objective
HDPE	High-Density Polyethylene	**SSC**	Sea State Code
HLO	High Level Objective	**WIML**	Wave-Induced Mismatch Losses
IEA	International Energy Agency	**WMO**	World Meteorological Organization
IEC	International Electrotechnical Commission		

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1 Excellence

Scaling up renewable energy (RE): rapidly and sustainably. Solar photovoltaic (PV) technology is widely recognized as key to realizing fully decarbonized energy generation and a main lever to achieve the European Green Deal objectives while phasing out our dependence on Russian fossil fuels. PV module installation rates have increased tremendously over the last ten years, driven by reduced costs of PV and increased climate awareness. We need to keep increasing the deployment rate for many years to come if we are to meet EU policy goals[1]. With increasing PV deployment, issues related to land access will become increasingly important. Integrating PV in architecture, infrastructure and agriculture are promising solutions to abate land scarcity. Floating PV (FPV) is another[2].

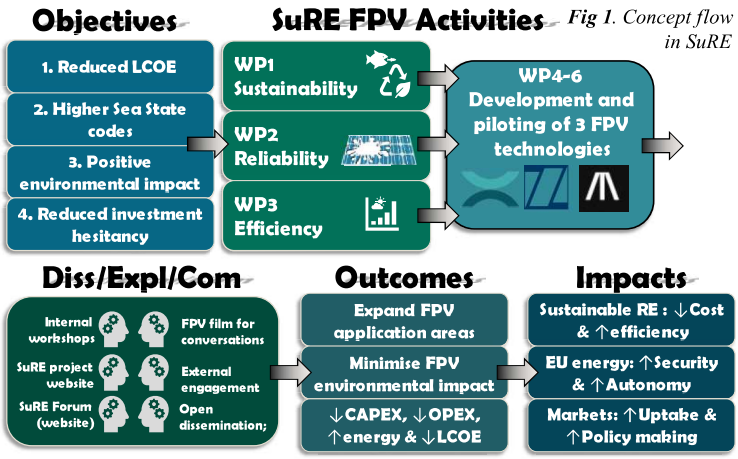


Fig 1. Concept flow in SuRE

Through **SuRE** we will enhance the affordability and sustainability of FPV. Enhanced **Su**stainability will be achieved by reducing the carbon footprints and enhance positive synergies with the local environment. The inherent modularity of PV provides an opportunity to tailor coverage ratios, light penetration, and layout to achieve FPV power plants with a positive impact on the local environment and biodiversity. We will achieve enhanced affordability by making more **R**eliable and cost-**E**fficient FPV power plants, developing models and framework that reduce the financial and reliability-related risks of this still young technology. Finally, we use existing and new knowledge to significantly improve the designs of three specific FPV technologies. These include the two most widely used FPV technologies in Europe: Ciel et Terre (CTI) and Zimmermann (ZIM). The third innovative FPV technology, Sunlit Sea (Sunlit), aims at offshore applications. All three technologies have separate and concrete routes laid out in SuRE to reduce LCOE through enhanced reliability and cost-efficiency, while reducing carbon footprints and enabling a positive impact on the local biodiversity. The activities to reach these goals are structured by entwining 3 generalizable work packages (WPs), ***WP1 Sustainability, WP2 Reliability, and WP3 Efficiency***, with 3 value-chain specific WPs focused on innovations for the 3 FPV technologies, ***WP4 CTI, WP5 ZIM, and WP6 Sunlit***, as illustrated in Fig. 1. This approach benefits both a broad European FPV industry *and* the individual FPV value chains in SuRE. Consequently, SuRE provides a clear pathway to substantially increase the deployment of FPV in Europe and expand the application of FPV on inland and offshore waters, in line with the EU strategic priorities in Table 1.1.

Table 1.1: Relation of SuRE to EU strategic priorities.

EU Strategic topic	*Contribution of SuRE*
EU Climate strategies and targets. 2050 long-term strategy	The EU requires decoupling of growth and CO_2 emissions. The transition to a climate neutral society is both an urgent challenge and opportunity to build a better future for all. This is at the heart of the European Green Deal. SuRE, with our commitments to accelerating economical and sustainable RE supports these aims.
The European Strategic Energy Technology Plan	The European Strategic Energy Technology Plan (SET Plan) aims to boost the transition towards a climate neutral energy system through the development of low-carbon technologies in a fast and cost-competitive way. The objectives of SuRE are perfectly aligned with the two first SET plan key actions ("N°1 in renewables").
REPower EU	The EU has published the plan REPower EU to rapidly reduce dependence on Russian fossil fuels and fast forward the green transition. As part of the REPower EU plan, the European Commission aims to bring online over 320 GW of PV by 2025 and almost 600 GW by 2030. SuRE can accelerate FPV rollout –responsibly– to be an important wedge of the PV pie to provide "a truly interconnected and resilient EU energy network."
The EU Solar Energy Strategy from the EU Commission	Amid the energy crisis and geopolitical tensions, the implementation of the Strategy and key solar initiatives proposed for the EU and its member states is of utmost urgency, as reflected in this document approved by the EU Commission in May 2022. SuRE addresses the caution that "required expansion in utility-scale projects will increasingly

	face competing uses of land and public acceptance challenges" by making FPV, a non-land based RE more cost competitive and sustainable.
The Paris Agreement	To keep mean global temperatures well below 2 °C above pre-industrial levels, decarbonization must accelerate. SuRE aims to contribute to this, while increasing society's ability to foster climate resilience by, for example, pairing PV production with water-reservoir stewardship and avoiding competition with food production lands.
The European Industrial Strategy	The Strategy seeks to strengthen the single market, continue openness to trade, and accelerate the green and digital transitions. SuRE encompasses several FPV suppliers, who are headquartered in Europe, deploy utility-scale FPV in Europe, source material and labour from Europe, and strengthen the EU's open strategic autonomy.

1.1 Objectives and ambition #@PRJ-OBJ-PO@#

1.1.1 Objectives and relation to the work programme

SuRE provides targeted answers and measurable solutions to the topic HORIZON-CL5-2023-D3-01-03: Floating PV Systems, addressing systematically the five objectives of the Scope. The consortium has set a robust implementation plan to ensure the success of SuRE and de-risk the project.

The overall SuRE objective is to increase competitiveness of FPV power plants compared to fossil fuels and land-consuming renewable technologies by enabling sustainable design, reducing LCOE and expanding the application potential for three leading European FPV technologies.

This objective of SuRE is based on long-term development plans of the three FPV technology providers in the consortium: 1) Ciel et Terre (**CTI**, France) was early to provide utility-scale FPV solutions. Since 2011, they have installed 650 MW globally, with 1.2 GW under development. Their subsidiary Laketricity (**LKY**, France), devoted to utility-scale FPV projects development, has delivered 90 projects in 10 countries. CTI and LKY will build a 5 MW FPV pilot plant situated on Lac de Tertre, Saint Ghislain in Belgium, using CTIs recently prototyped floater design. This new design is a radical departure from their existing Hydrelio technology. It provides reduced costs, increased sustainability, and expanded waterbodies for installation. 2) Zimmermann PV-Steel Group (**ZIM**, Germany) and their long-term developer partner, BayWa r.e., (**BayWa**, Germany) have installed 210 MW of FPV, including the Sellingen project in the Netherlands which is the largest FPV system outside of Asia. ZIM and BayWa will build a 5 MW FPV pilot in the Netherlands, to extend their FPV design for more waterbodies while advancing on all SuRE fronts. BayWa has a pipeline of more than 2 GW of FPV projects. 3) Sunlit Sea (**Sunlit**, Norway) has developed a floater-module integrated FPV system. They have several pilot projects in Norwegian and US waters. **Metsolar** (Lithuania) makes custom PV modules and has worked with Sunlit for years. Sunlit will update their design to decrease resource burden and production cost by using SuRE to bring new technologies into their production line. During the project, SuRE will build a 105 kW pilot at Singløya, Norway. Augmenting these 3 independent FPV value chains, SuRE includes **SABIC** (Netherlands) as a technical innovator of bio- and circular plastics for the FPV market. SABIC will work across the value chains and with the four independent research institutes to deliver plastic sustainably. Four research institutes and one science communication association complete the SuRE consortium: **IFE** (Norway) coordinates SuRE, **Fraunhofer** (Germany) is the largest solar research institute in Europe, **TNO** (Netherlands) is an independent research association for targeted innovation, **Deltares** (Netherlands) is an independent knowledge institute in water and subsurface. All four institutes have been forerunners of driving FPV science and technology. **Compáz** (Switzerland) is a collective of visionary scientists and artists who push social progress in Europe. Together the SuRE consortium is exceptionally well positioned to impact the future of FPV technology and enable a more rapid deployment. SuRE will expand the application areas, minimize environmental impact, and decrease LCOE to make FPV a robust source of RE.

*There is no clear classification of FPV technologies for application on inland and/or offshore waters. To specify wave conditions the FPV technologies are designed for, we use the Sea State Codes (SSC) of the World Meteorological Organization's (WMO). The scale spans from wave height 0 m (SSC 0) to wave height > 14 m (SSC 9). The term "**inland**" FPV will be used for deployment of FPV on **freshwater** bodies, while the term "**offshore**" FPV will be used for deployment of FPV at **sea**. The design-SSC must be evaluated for the individual location/water body.*

To achieve our overall objective, we have defined the High-Level Objectives (HLO) and Specific Objectives (SO) in Table 1.2. Key Performance Indicators (KPIs) describe how SOs will be measured and verified.

Table 1.2: High-Level call Objectives and corresponding Specific Objectives in SuRE.

HLO1. Reduce LCOE for FPV technologies and verify technology on adequate scale (minimum 5 MW)
SO1:Improve design and reduce material consumption to decrease CAPEX for three European FPV technologies
How SuRE achieves it: 1.1a) Develop a new FPV technology patented by CTI, including a 5 MW pilot. The 5 MW pilot, finished in M18 of SuRE, will be used as a basis for further testing and improvements to the technology. The core of the development is a shift in O&M strategy that will reduce the Balance of System (BoS) cost. BoS costs is the main factor that drives LCOE for FPV to be higher than for Ground-mounted PV (GPV) **1.1b)** Reduce BoS costs for ZIM FPV technology through new schemes for interconnects and mooring, and more accurate models for the forces on the structure to avoid overengineering of the steel components. Developments will be verified in a 5 MW pilot, starting construction in M18. **1.1c)** Develop a tool for multi-dimensional optimization of Sunlit's patented floating structure, with respect to cost, efficiency, and structural robustness. **1.1d)** Collaboration among the strong value chains represented in SuRE will contribute to transferability across sites, water body types, and manufacturers. **KPIs:** CTI targets an LCOE reduction of 10% compared to their Hydrelio technology. ZIM targets a 15% reduction in Structural BoS. Sunlit has calculated a potential LCOE reduction of up to 60%, due to high potential for reductions in CAPEX. The exact reduction depends mainly on the optimization of the aluminum pressing. **WPs:** 2, 3, 4, 5, 6
SO2: Improved FPV designs and O&M tools to decrease OPEX for all FPV technologies
How SuRE achieves it: 1.2a) CTI is developing a new technology design which will allow lowering of O&M cost. An associated new O&M method will be developed in the project **1.2.b)** IFE will develop algorithm for endogenous determination of optimal cleaning intervals. **1.2c)** The algorithm for optimal cleaning intervals will be tested on existing power plants from the different consortium participants **1.2d)** Existing O&M routines will be adapted to be applicable at higher SSC. **1.2d)** Fraunhofer and IFE will develop a digital twin for the CTI technology to enable more efficient fault detection and maintenance routines. **1.2.e)** Collaborative R&D among commercial partners and research institutes in SuRE will contribute to establish and publish pre-normative practices. **KPIs:** New CTI O&M strategy implemented on pilot. Best practice O&M for FPV published in International Energy Agency (IEA) Photovoltaic Power Systems Programme (PVPS) report. Algorithm for Optimal Cleaning Intervals for FPV published. Open access publication of algorithm for cleaning optimization for FPV. **WPs:** 3, 4, 5, 6
SO3: Develop tools to improve yield for all FPV technologies and apply the tools for concrete optimization of the 3 FPV technologies in SuRE
How SuRE achieves it: **1.3a)** IFE and Fraunhofer will develop and verify a model for calculating the operating temperature of PV modules in FPV applications that accounts for all the different cooling mechanisms (evaporative, conductive, convective, and radiative). **1.3b)** The complete thermal model will be used together with the FPV technology providers to optimize the designs with respect to cooling. **1.3c)** TNO and IFE will develop a new model for wave induced mismatch losses (WIML). **KPIs:** A significant reduction of thermal losses of ~3% for CTIs new technology compared to Hydrelio. A reduction in thermal losses for Sunlit technology of ~3% (6 degrees reduction in operating temperature). Quantified and verified U-values for the 3 FPV technologies. Open access publication on thermal model for FPV power plants. Guidelines for predictive yield modelling of FPV in PVsyst **WPs:** 3, 4, 5, 6.
HLO2. Ensure reliable operation and extend the application of low-cost FPV technologies to offshore conditions
SO4: Demonstrate system components that satisfy structural and functional requirements for the entire lifecycle.
How SuRE achieves it: 2.1a) Expand ZIM Float design standards from Sea State Code 3 (SSC 3) WMO - *Slight*, wave height 0.5 – 1.25 m) to SSC 4 (*Moderate*, wave height 1.25 - 2.5 m) through optimization and increased structural integrity of their steel components, and development of new mooring solutions. **2.1b)** Perform environmental stress fatigue crack testing of floater interconnects **2.1c)** Test and verify steel coatings applicable in higher salinity water. **2.1d)** Validate Sunlit design up to SSC 5 (*Rough*, 2.5 – 4 m) with respect to structural integrity and moisture ingress **2.1e)** Develop and verify plastics components with satisfactory UV stability. **2.1e)** Analysis of reliability challenges resulting from soiling and fouling and potential mitigation measures at module level. **2.1f)** Analyze expected increased loads on mooring and floating structures due to algae growth. **2.1g)** Analyze performance loss rates (PLR) from new pilots and existing systems. **KPIs:** New design standard for ZIM Float for SSC4. Validation of Sunlit technology for SSC5. Report on corrosion of steel components. Report on UV stability of plastic components. Report on PLR. Quantify soiling-induced hot spot loss in 2 pilots. **WPs:** 2, 4, 5, 6
SO5: Accelerated stress testing procedures for FPV components
How SuRE achieves it: 2.2a) Develop new accelerated stress tests designed explicitly for FPV by combining degradation mode statistics, quantified field stress levels, and insights from modelling for three key stressors/degradation. **KPIs:** Publication and revision of pre-normative procedures. Adoption of, engagement with, and refinement suggestions of procedures by FPV producers and installers. **WPs:** 2
HLO3. Demonstrate low impact on biodiversity and satisfy end-of-life requirements

SO6: Develop realistic targets, measurements, and monitoring to assess FPV sustainability
How SuRE achieves it: 3.1a) Fraunhofer and Deltares will develop and validate monitoring and modelling methodologies for hydrological and ambient parameters relevant for assessing the interaction of FPV with waterbodies and ecosystems. **3.1b)** Use CTI and ZIM 5 MW pilot plants as monitoring and measurement testbeds. **3.1c)** Simulation tool-chain for modeling the impact of FPV plant coverage on waterbody (oxygen levels, temperature change and others). **KPIs:** Pilot plants outfitted with extra measurement and monitoring equipment and routines. Sustainability data input from FPV plants other than pilots. Simulation tool-chain extended and verified by measurements. SuRE technology developers/off takers meet or exceed the targets developed in SuRE for positive environmental impact. Adoption or trailing of targets by non-SuRE commercial actor. **WPs:** 1, 4, 5
SO8: Guidelines for FPV Sustainability and potential mitigation measures
How SuRE achieves it: 3.2a) Results from 3.1), combined with identified pathways to environmental positive FPV and FPV designs that contribute to enhance biodiversity will be used to develop the sustainability report with pre-normative guideline to assess the direct impacts of FPV on aquatic systems, including on water quality and toxicity, biodiversity, potential mitigation measures and circularity. **3.2b)** If negative impact is found, potential mitigation measures will be suggested and included in the guidelines. **3.2c)** Further selected FPV plants available to the consortium will be used for broad-scale testing to ensure EU-wide applicability. **3.2d)** Measurements of water quality and toxicity will be carried out for 10 of CTI's existing FPV sites. **KPIs:** Openly published sustainability report with guidelines to assess the direct impacts of FPV on aquatic systems and biodiversity, including potential mitigation measures. **WPs:**1, 4, 5, 6
SO9: End-of-life recycling aspects and reduced carbon footprints for FPV components
How SuRE achieves it: 3.3a) CTI and SABIC will work to increase the use of recycled materials in the production processes of the new CTI technology. They will also explore alternative materials, moving from post-industrial recycled HDPE to post-consumer recycled HDPE or PP, with lower carbon footprints. **3.3b)** ZIM technology will have reduced material consumption because of more accurate engineering for different SSC. **3.3c)** Sunlit will explore if recycled and/or thinner Al can fulfil the requirements of their floater using the new knowledge on Al requirements from detailed modelling of aluminum pressing. **3.2d)** Ensure that the components satisfy end-of-life recycling aspects. **KPIs:** ZIM aims to reduce material consumption and CO_2 footprint of ZIMFloat by 20%. CTI aims to move from post-industrial to post-consumer recycled material. Sunlit aims to cut Al material consumption by 50% and evaluate use of recycled Al. Documented LCAs and end-of-life recycling aspects for each technology. **WPs:**1,4,5,6
HLO4. Reduce hesitancy to invest in FPV power plants
SO9: Enable accurate prediction of LCOE for FPV Power Plants
How SuRE achieves it: 4.1a) Develop add-ins and tables to improve accuracy of energy yield assessments (EYA) for FPV power plants using existing, bankable PV modelling tools. **4.1b)** Streamline LCOE calculations to accelerate investment process, also including propagating uncertainty analysis. **4.1c)** Identify the most sensitive parameters through Pareto front analysis for historical and forthcoming FPV plants. **KPIs:** Publications on detailed EYA inputs for FPV in PVsyst. Sensitivity analysis of FPV plant decisions on LCOE. **WPs:** 3, 4, 5, 6
Specific Objective 10: Reduce "soft" barriers for deployment of FPV power plants
How SuRE achieves it: 4.2a) Facilitate communication between the public and FPV providers through documentary film, social media platforms, and hearing sessions with societal stakeholders. **KPIs:** Attendance and engagement with hearing sessions, social media engagement, distribution of documentaries. **WPs:** 7, 8

1.1.2 Ambition and Progress beyond State of the Art

Sustainability guidelines Currently, an FPV project will usually require an environmental permit (or equivalent). Identification of impacts the FPV project might have in and around the deployment location is done through an environmental impact assessment (EIA). The EIA should include impact on i) flora and fauna ii) habitats iii) noise iv) visual v) soil, and vi) water quality and human health. To support such assessments, however, existing literature on local environmental impacts of FPV systems is limited, with mostly suggestive and unsupported findings, and significant methodological limitations. *SuRE will go beyond this state-of-the-art* by establishing a set of sustainability evaluation metrics and a sustainability monitoring methodology for FPV and employing this at SuRE-demonstrator and -associated FPV sites. The results will be used to validate models, which will then be used to map out combinations of FPV technologies and European waterbody-types relevant for FPV. Further, SuRE will engage with stakeholders to integrate nature-inclusive principles and climate-adaptivity in FPV design, not currently an integrated part of the design principles. The research will lead to guidelines for FPV designs and deployment to minimize potential negative impacts and enhance positive impacts on biodiversity and climate adaptivity.

Carbon footprint and life-cycle impact of FPV technology. In addition to local impacts at the deployment location, the production of the components making up the FPV system also has sustainability implications. Preliminary analysis by TNO indicate that besides the PV modules, the floaters currently dominate the CO_2 footprint. Further, the analysis showcased how a framework to fully assess the life cycle impacts of FPV is lacking. *SuRE will go beyond this state-of-the-art* by reducing CO_2-footprint of the three SuRE FPV technologies. The project developments will allow significantly reduced material consumption of plastics (CTI), steel (ZIM) and aluminium (Sunlit) for the floaters, as well as sourcing of plastics and aluminium from lower grade (e.g., plastics from bio-based monomers) and/or recycled material. Further, SuRE will develop an FPV-specific LCA framework, incorporating both global impacts from the production of FPV components, as well as local impacts on e.g., ecosystems and water quality.

FPV specific accelerated stress tests. Commercially available PV panels and components are developed based on GPV experience, as is existing qualification and safety test standards (e.g., IEC 61215, IEC 61730). New risks specific to FPV calls for adapted or new standards to be developed, due to altered stress profiles for mechanical loads, humidity, salt exposure and fouling. Tests to go into such standards have yet to be established, validated, or adopted by the market. *SuRE will go beyond this state-of-the-art* by building required scientific basis of FPV-specific degradation mode statistics and stress profiles. This will be combined with development of modelling tools to allow transferability and extrapolation between technologies and SSCs. On this basis, FPV specific accelerated stress tests will be developed. These will contribute to establish recommended practices and pre-normative protocols, moving the community towards dedicated FPV qualification standards.

Accurate EYA modelling tool for FPV. State-of-the art EYA modelling PV tools, such as PVsyst, SAM and HelioScope do not address FPV-specific issues that affect the yield and performance of FPV systems. Lacking model concepts include i) influence of all relevant parameters for module temperature calculations, ii) influence of WIML, and iii) appropriate values for soiling losses and PLR/system degradation. These introduce significant uncertainty in EYA- and consequently LCOE-estimates, lowering investor appetite and hampering targeted technology innovation. In addition, standard LCOE schemes assume small standard deviations of the input variables, which need not be the case. *SuRE will go beyond this state-of-the-art* by developing new, accurate and FPV-specific modelling tools that can be used as add-ons for existing bankable EYA tools for PV power plants, such as PVsyst. We will also develop adapted procedures to obtain and establish recommended values for heat loss coefficients, WIML, soiling losses and PLR for FPV systems. This will improve LCOE calculations in general, and those used specifically to evaluate SuRE's technological innovations. The LCOE calculations will further use enhanced uncertainty propagation methods for LCOE calculations, to calculate probabilities for profitability. Lastly, the improved yield models will be input to a digital twin for the CTI pilot. Implemented in the asset management, the digital twin can improve the predictive capacity of the monitoring software relative to today's commercial solutions and hence reduce O&M costs.

Redesign of floaters and anchoring to reduce CAPEX/SBOS and increase structural integrity. LCOE of FPV power plants is generally 12-25% higher than for GPV, mainly from added SBoS costs of floaters. In addition, low-CAPEX FPV technologies need to increase their structural integrity to adapt to higher SSCs and thereby expand the FPV market. *SuRE will go beyond this state-of-the-art* by developing an FPV-specific modelling chain for mechanical loads, combining advanced hydrodynamic, computational fluid dynamics and component finite-element modelling (FEM). This will allow proper optimization of system and component structural integrity while reducing net material consumption for floaters and anchoring by avoiding over-engineering, for all SuRE FPV technologies.

New FPV technology from CTI. CTI's established FPV technology, Hydrelio, uses standard PV modules mounted on blow-moulded HDPE on plastic floaters. This CTI design is Europe's most-installed FPV technology. *SuRE will go beyond this state-of-the-art* by developing and demonstrating a novel CTI design, radically new in both installation and layout (patent pending, FR310956, filed in France in 2020). The design decreases carbon footprint and LCOE. SuRE will support this technology development both through general and technology-specific innovations. This includes the development of a new maintenance device to match the new design and significantly reduce cost.

Optimization and standardization of ZIM's FPV technology for higher SSCs. ZIM uses standard PV modules mounted on steel structures, which are embedded in HDPE floaters. This technology is used in Europe's largest FPV plant of 41 MW, 76 000 modules. ZIM and BayWa currently have standard design practices for WMO SSC 3 (wave height 0.5-1.25 m). Due to lack of measurement data and modelling tools, however, projects at higher SSC are done on case-by-case basis and have the steel structure and mooring system systematically overdesigned, to ensure integrity. *SuRE will go beyond this state-of-the-art* by making measurements, model- and design-improvements, and testing of new structural components to optimize and standardize the structure for rougher sea states. This will expand ZIM's markets, drive down cost, and lower material-use by avoiding over-engineering.

Digital optimization cycle for Sunlit's floater design. Sunlit applied for two patents in 2019, has a production facility in Askim, Norway, has built 10 prototypes for R&D, and recently shipped its first commercial delivery. Sunlit integrates the module directly on top of an aluminum floater. It is an FPV solution that is robust for many waters, allow for a high degree of prefabrication, has good thermal contact with the water and is rapid to deploy. However, the current floater design is sub-optimized, due to a time-consuming and expensive optimization cycle. *SuRE will go beyond this state-of-the-art* by coupling the aforementioned mechanical modelling-chain to a SuRE-developed modelling tool for their aluminum press process and to heat transfer modelling, to allow dramatically accelerated multi-dimensional digital optimization of Sunlit's floater-design.

1.1.3 Project Positioning in terms of R&I maturity

Main activities in SuRE cover developing and piloting new technologies for use in FPV power plants. Developments span from novel concepts and products by Europe's largest FPV technology providers to the introduction of new materials, approaches, and design modifications to radically enhance existing FPV components and products. We bring together Europe's leading commercial and research expertise within FPV. Through SuRE, FPV deployment will increase as we provide solutions to overcome today's key barriers for increased FPV adoption: 1) documentation of, and guidelines for, sustainable deployment on a broad set of waterbodies; 2) enhanced structural integrity for low-cost FPV technologies to extend to offshore conditions; and 3) reduced LCOE to increase competitiveness against other, land-consuming or polluting, electricity production. Table 1.3 summarizes SuRE's TRL advances for key technological components. Specific technical activities to reach these TRLs are given in the indicated WPs.

*Table 1.3: Innovation Advancement Ladder in SuRE (*TRLi, TRL_f: Initial and Final Technology Readiness Level).*

INNOVATION ADVANCEMENT LADDER IN SuRE	
Innovation	**Current limitations, Partners, TRL_i/TRL_f and involved WPs**
Sustainability Guidelines to support better environmental impact assessments for FPV	Current limitation: Unclear or vague targets for sustainability in FPV and the impact of coverage on water quality. Insufficient knowledge base to carry out ESIA for FPV. Sustainability often a concern in policy and permitting. Partners: ALL, TRL3/TRL6, WPs:1,4,5,6
Enhanced use of recycled plastic for FPV components	Current limitation: Production of floater structures are carbon intensive. Partners: CTI, SABIC, TRL4/TRL6, WPs: 1, 4, 5, 6
Pre-normative accelerating stress tests	Current limitations: Accelerated stress tests exists only for GPV, no standards are yet set for FPV specific stress profiles. Partners: TNO, IFE, Fraunhofer, CTI, LKY, ZIM, BayWa, Sunlit, Metsolar, TRL5/TRL6, WP: 2
Accurate EYA modelling tool for FPV	Current limitation: State-of-the art tools used to assess bankability and LCOE from FPV systems cannot address FPV specific factors. Partners: ALL, TRL3/TRL6 WPs: 2, 3
Optimization of anchoring systems based on more detailed simulation of loads	Current limitation: Anchoring systems are typically dimensioned for worst case scenarios, driving up CAPEX and LCOE. Partners: CTI, LKY, ZIM, BayWa, Sunlit, IFE, TNO, TRL4/TRL6, WPs: 4, 5, 6, 2
Redesign ZIM Float SBoS and connectors to increase structural integrity and reduce CAPEX.	Current limitation: LCOE of FPV generally 15-25% > than for GPV. Expand today's low-cost, inland FPV solutions to rougher environments. Partners: ZIM, BayWa, TNO, IFE, Fraunhofer, TRL3/TRL6, WPs: 4, 2, 3
New CTI technology designed to reduce CAPEX	Current limitation: LCOE of FPV power plants generally 15-25% higher than for GPV. Partners: CTI, LKY, IFE, TNO, Fraunhofer, $TRL4/TRL_{pilot}5$/TRL6. WPs: 4, 2, 3
Develop new maintenance device, removing the need of walkways, to be tested on the Pilot	Current limitation: Walkways are used to provide access to FPV systems for O&M but are driving up CAPEX. Partners: CTI, LKY, TRL3/TRL7, WP: 4
Mitigation measures identified in WP2 realized in an industrial production line at Metsolar	Current limitation: Production of PV modules with adaptions for rougher FPV applications is not undertaken due to economics of scale Partners: Metsolar, TRL2/TRL6, WP2
Integrated model-chain for Sunlit floater optimization (Al pressing, heat transfer & structural integr.)	Current limitation: Inconvenient and expensive procedure for design improvements of Al float. Partners: Sunlit, IFE. TRL2/TRL6, WPs: 6, 1, 2, 3

1.2 Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

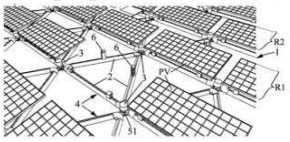
SuRE achieves its objectives by innovating along two main lines: general developments beneficial for the whole industry, and specific developments to enhance the 3 individual FPV technologies of SuRE. The project methodology will in the following be presented under the headlines of Sustainability, Reliability and Efficiency. In each of these sub-sections, the general methodology will be presented first, and then the technology-specific approaches for each of the value chains: CTI (with LKY), ZIM (with BayWa), and Sunlit (with Metsolar). Each technology developer is making improvements to their own FPV technology, central to reach the companies' expansion targets, as well as the overall SuRE objectives. The innovations and associated pilots to be built in the project are described below.

Overview of the technology innovations and associated pilots

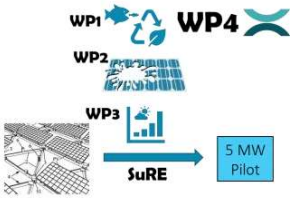
The overall goal of CTI in SuRE is to develop their innovative design. The new layout reduces ground transport to site, while also increasing material and surface efficiency for light harvesting by reducing maintenance gangways. Developments of this technology, to keep in line with CTI and LKY practices, requires co-optimization of material sustainability, O&M practices, ecologically beneficial spaces, FPV reliability, and LCOE reduction. **CTI and LKY will build a 5 MW pilot** during the first 18 months of SuRE. The pilot site is selected, and CTI's new design is ready for rollout. A first prototype island of 150 panels has been built and is floating at CTI's test site. A smaller 50 kW pre-pilot will be built immediately after project initiation, to prepare for the larger pilot. The 5 MW pilot will cover at least 5 ha of the 15.3 ha waterbody, using minimum 8000 modules of 650 W. Pilot sensorics will provide data also beyond the project period. Planning permission for the site was in January 2023.



Fig. 2. CTI will pilot the new design, making use of progress in WPs 1-3.



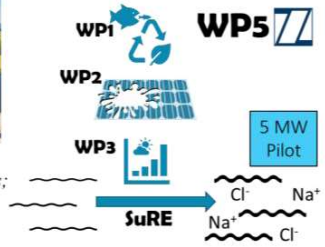
CTI's new design to be tested in SuRE is patent pending. The image above is Fig 7 from Patent FR310956 (2020).



The overall goal of ZIM in SuRE is to expand their market by enabling deployments in rougher conditions. Their present large-scale FPV installations are mostly on fresh inland waters like water reservoirs and quarry ponds; in corrosion category C3 (medium) and with significant wave heights under 1 m, thus SSC 1-3. Components and standard design practices will be improved and updated based on measurements and model improvements in SuRE to move to SSC 4, also allowing for more corrosive environments. **ZIM and BayWa will build a 5 MW pilot** plant in the second half of SuRE to put these new designs and components to the test. The pilot will incorporate up-to-date developments specific to the ZIM Float technology and the generalizable Sustainability, Reliability, and Efficiency activities. The pilot site in the Netherlands will also be used for pre-, mid-, and post-install studies for water quality, shoreline effects and ecology. Once built, the pilot is a platform with a commitment to future access for study. With the pilot constructed in the middle of the project period, we will report on one annual cycle of operation before project end.



SuRE FPV allows Zim/BayWa to push the boundaries of its services; key interactions are with WP1-3.



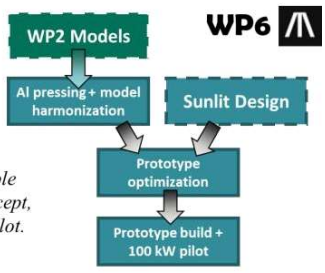
The overall goal of Sunlit in SuRE is to optimize their floater design to unlock significant improvements in LCOE and sustainability, while maintaining a sturdy design. In Sunlit's design two aluminum sheets pressed into an egg-carton-like design face each other and trap air to make a pontoon; the PV module is affixed directly to the top aluminum sheet. Heat is efficiently transferred via the metal and into the water. Neighboring floats are connected to each other with controllable hinging, in accordance with expected waves. Sunlit's design has received preliminary DNV verification. The present Sunlit design uses partially optimized aluminum pressing. Key improvements pursued in SuRE are reduced aluminum consumption, increased digital prototyping to reduce time-to-prototype construction, and increased production throughput and automation. **Sunlit and Metsolar will build a 100 kW-pilot** during SuRE, with extra sensors and monitoring to support the generalizable results, also beyond the project period. The pilot will be deployed on the coast of Norway.



***Fig 4** The integrated pontoon-PV module is a modular, robust design with efficient water cooling. Pontoon inter-connections designed to waterbody needs.*



SuRE FPV combines generalizable methods with existing Sunlit concept, ending in a new prototype and pilot.



1.2.1 Towards Sustainable, Reliable and Efficient FPV Power plants

1.2.1.1 Sustainable FPV power plants (WP1 + WP4-6)

Challenges and opportunities related to FPV sustainability can be broadly categorized into (1) local impacts, resulting from the FPV system deployment at the site, and (2) global impacts, from producing components that go into the FPV system. For each of these, impact must be evaluated within three dimensions of sustainability - economic, social, and environmental. SuRE will provide a comprehensive scientific knowledge base looking at all these dimensions and their interaction, particularly focusing on the environmental impact of FPV installations.

Assessing the local environmental impact of FPV deployment. As given in Section 1.1.2, there are severe limitations in the knowledge base underpinning the environmental impact assessments (EIAs) required prior to the building of an FPV plant. In SuRE, we will fill these knowledge gaps by combining monitoring and modelling.

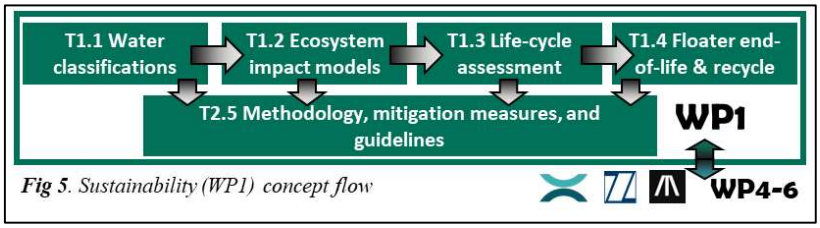


Fig 5. Sustainability (WP1) concept flow

Monitoring data (in-situ) of the key parameters of the water body, food web and the ecosystems are needed. The food web describes who eats whom in an ecological community. Food webs help us understand how changes to ecosystems affect many different species. Anticipated effects of FPV include the limitation of sunlight penetration into the water column and changes in water temperatures. In this way FPV affects the foundation of food webs, namely photosynthesis; the production of food and oxygen. We therefore focus our monitoring efforts on light penetration, oxygen, temperature, nutrients, and the production of phytoplankton. Today, experience on measuring these parameters under and around an FPV system is limited. Therefore, SuRE will develop a monitoring strategy for the project demonstrators to capture any relevant effect under and around an FPV system both in the temporal and spatial domain. This monitoring strategy will be used to gather relevant data from the project pilots and existing SuRE-associated plants, as input to sustainability guidelines and modelling. The monitoring data will be complemented by gathered system knowledge, a term referring to information on the waterbodies and surrounding landscape. Examples of such data is water body depth, nutrient loads, currents, water exchange, type of agriculture in the vicinity etc. Such knowledge is crucial to understand the processes leading to changes in water quality and biodiversity.

A major impact of FPV installations on waterbodies is the latter's change of energy balance through the blockage of light. In waterbodies, this triggers changes in thermal stratification, cycles of chemical compounds and food-webs, affecting biodiversity below and eventually also above the water surface. In this project, the dependency of light penetration on several FPV designs will be modelled through standard tools that are mainly based on raytracing algorithms, namely Delft3D, BIGEYE and a RADIANCE-based module in the yield simulation tool-chain ZENIT.

Alongside data gathering and light simulations, hydrological modelling is needed to forecast consequences for biodiversity (algae, submerged vegetation, mussels, fish, and birds) under a varied number of technological conditions in different waterbodies. To this end, two complementary modeling frameworks will be used: First, the Fraunhofer FPV-GLM, an enhanced General Lake Model, will be used with its biogeochemical aquatic ecodynamics (AED2) submodel. This is a fast 1-D model, suitable to explore a wide and representative range of European waterbodies in different climate zones. As a function of the water coverage ratio and resulting ratio of blocked irradiation, the model will be used to map out each waterbody's energy balance. This will result in estimates of the maximum ratio of blocked irradiation beyond which water quality and biodiversity criteria given by the Water Framework Directive 2000/60/EV are violated. Further, the expected evaporation reduction will be recorded. This exploratory work focusing on a broad selection of waterbodies will be validated and complemented by more detailed (temporal and spatial) modelling with Delft3D. This model captures the impact of FPV systems on waterbody hydrodynamics and temperature in three spatial dimensions. It will be employed for a smaller subset of typical waterbodies, including those with SuRE demonstrators or SuRE-associated FPV plants already deployed. In the neighborhood of values approximated by the FPV-GLM, maximum ratios of blocked radiation will be evaluated in a much more detailed manner by Delft3D. Similarly, computed functional dependencies of evaporation rates on coverage ratios and FPV designs will be compared across model frameworks used by Fraunhofer, Deltares and IFE. Calibration (when necessary) as well as validation of both models will be achieved through the aforementioned monitoring data from demonstrators. Additional to an improved understanding of the applicability of each modelling tool, including error evaluation, this will result in a catalogue of maximum permissible FPV coverage ratios of European waterbodies differentiated by waterbody type and different FPV designs.

The potential of positive local environmental impact from FPV deployment. SuRE strives beyond minimizing negative local impacts and explores possibilities to achieve positive local impacts of the FPV system. This will be done by enhancing the integration of nature inclusive and climate adaptive principles in FPV system design. In this process, proper involvement of and interaction between stakeholders will be crucial. A range of different methodologies exists for facilitating and supporting participation and stakeholder processes. Within the setting of this project, we will use a specific method of action research: group model building (GMB). GMB is a visual and practically useful method for stakeholder participation in water management to develop and improve decision support models as well as integrate existing information with local/stakeholder knowledge. Using GMB we will organize design sessions on the SuRE demonstration sites with technicians, landscape designers and ecologists, where various designs will be evaluated. Focus will be on achieving a joint definition of barriers and enablers for achieving sustainable FPV system designs, encompassing both the social, economic, and environmental dimensions. Further, a set of sustainable design examples will be developed. Ultimately, we will create and share guidelines for development and implementation of FPV solutions with a minimally negative and/or maximally positive environmental impact.

Sustainability impacts of the FPV system components and materials; a global LCA perspective including circularity. In addition to the study of local impacts at the deployment site, SuRE will use life-cycle assessment methodology to quantify material use, energy use, and related emissions during product and service life cycle. SuRE will compute, for all components of the three SuRE technologies, the CO_2 footprint and other environmental footprints using procedures like EF 3.0 or ReCiPe. This work will serve to guide and quantitatively assess changes in design and material selection, and to evaluate end-of-life recycling options, which will be further elaborated below.

To advance state-of-the-art, SuRE will also develop a more comprehensive FPV-specific LCA framework. The LCA will build on data from sustainability analyses of SuRE waterbodies, including metrics and impact categories that quantify effects of FPV installations on waterbody biodiversity. A first approach has been tentatively explored for FPV[3] by addressing ecosystem damage through relative biodiversity loss. In SuRE, we will extend the LCA biodiversity metrics to provide a framework for holistic, cradle-to-grave LCA comparisons of different FPV designs.

Technology specific design changes, material selections and recycling possibilities. Plastics are a major part of the system for the FPV-technologies of CTI and ZIM, as well as for many other major FPV-technologies, through the high-density polyethylene (HDPE) buoys providing the system's buoyancy. SuRE brings together a plastics producer, FPV technology designers, FPV system suppliers, and research institutes to tackle a number of challenges related to this floater plastic. SuRE will assess the possibility of shifting production of the floaters from off-site blow molding to on-site injection molding. This would remove the need to transport excessive volumes (air trapped within the floaters) to site and opens the possibility to use polypropylene (PP). CO_2 footprint, and material and energy use are lower for PP compared to HDPE. Further, material sourcing from bio-based monomers and/or recycled polymer will be pursued. In the assessment and design of these solutions, SABIC will use their extensive experience and toolbox in predictive engineering for sustainable plastics. Combined with close collaboration between the partners along the value chain, this will ensure a comprehensive evaluation of the viability and proper implementation of the proposed solutions. In this work, circularity-by-design will be an inherent part of the consortium's approach.

For Sunlit's FPV-technology on the other hand, aluminum floaters provide buoyancy and is a major contributor to environmental footprint. Sunlit will use tools developed in SuRE to pursue the use of lower grade and lower volumes of aluminum. Such changes are expected to give substantially lower carbon emissions, which will be assessed by the LCA approach. A similar exercise will be performed on ZIM's technology, where advances achieved in SuRE will allow more exact and location-targeted dimensioning of the steel structures, decreasing material consumption.

1.2.1.2 Reliable FPV power plants (WP2 + WP4-6)

The historically established reliability learning cycle has served the PV industry well. It consists of three over-arching steps: (1) Field deployment of PV systems, (2) Failure- and degradation-mode characterization and analysis, and (3) Development of accelerated stress tests in the lab to screen rapidly for these failure and degradation modes[4]. In the face of reliability issues present in an FPV configuration, however, the established learning cycle is challenged in multiple ways. These will be described in the following paragraphs, alongside a description of the approach SuRE will employ to tackle these methodological challenges.

Statistics on failure and degradation modes. Open information on failure and degradation mode statistics from FPV sites is notoriously hard to come by, effectively inhibiting progress through the existing learning cycle. Although studies of material degradation in maritime environments are available, these do not focus on the exact materials and technology options of state-of-the-art PV. Consortium partners have previously collected incidental observations on FPV degradation modes, as summarized in Table 1.4. Building on this, SuRE will perform targeted campaigns to

establish the first statistically relevant mapping of degradation modes in the FPV configuration.

To this end, we use a combination of aerial infrared thermography (aIRT), statistically sampled field characterization through current-voltage (IV) scans and visual inspection, and root-cause analysis of selected modules in the laboratory. We will use both commercial and research systems available to the consortium. To speed up the process of reaching relevant statistics, time-series analysis on string level will be combined with aIRT to efficiently detect and quantify effects of various degradation modes[5].

Table 1.4 Collection of incidentally observed degradation and failure modes in FPV

Circumstance specific for FPV	System or component	Failure modes
Mechanical loads from waves, wind, water currents, and water-level variations	System	Lost mechanical system- and/or mooring- integrity
	Plastic components	Rubbing damage
	Plastic&metal components	Mechanical-stress induced damage
	PV panels	Glass cracking or deformation; PV cell cracking
	Float interconnectors	Accelerated materials fatigue, connectors breakage
	Inverter	Internal electrical disconnect ("electrical open") induced by continuous movements
	Cables	Acc. wear of cable mantles induced by continuous movements
Humidity and corrosion	Steel and aluminum parts	Accelerated corrosion
	PV panels	Moisture ingress in the panel; accelerated pot. ind. degradation
	Inverter	Corrosion induced defects
	Electrical system	Enlarged leakage currents, inverter shutdown
Pollution and fouling	PV panels	Hot spot ind. failures: diode failures, burn marks, microcracks
	PV panels	Accelerated output loss from (in)organic fouling

FPV specific stress profiles. Stress tests in the traditional learning cycle are, by design, highly empirical and lack well-established links to true stress levels. Instead, the tests are merely a means of ensuring that components meet a certain minimum quality level, in accordance with historical precedent. When module manufacturers still provide 25- and 30-year warranties (partly) based on these tests, it is guided by prior evidence that such lifetimes can be achieved by modules meeting this minimum quality level. The experience comes from GPV systems and traditional module architectures, however, and it is not clear how transferable this is to FPV. Further, only relying on degradation mode mappings of young systems (which is the only thing available for FPV) when developing accelerated stress tests can miss degradation modes that become prominent later in the plant's lifetime. Indeed, stressors that are expected to be more severe for FPV compared to GPV, such as humidity and repeated mechanical loads, are important for mid-life and wear-out degradation modes such as delamination, corrosion, or fatigue breakage. Thus, assessing applicability of current test frameworks, both on short- and long-term, requires knowledge about actual stress levels, which is currently lacking for FPV. SuRE will map out the profiles of key stressors for selected floating PV technologies by deploying targeted sensors at selected sites. These profiles will be compared to stress profiles for GPV[6–8] to assess where the largest differences are; hence the potential shortcomings of the current test framework. A special focus will be on the implications for mid-life and late-life degradation.

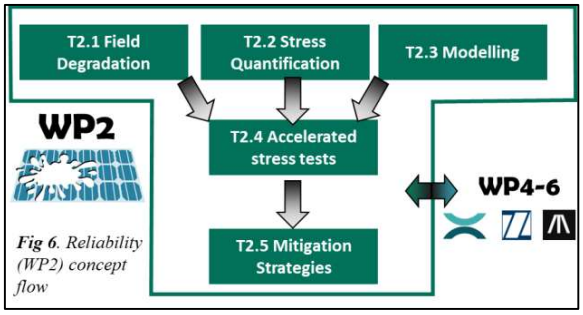


Fig 6. Reliability (WP2) concept flow

Accelerating the reliability learning cycle by help of modelling. The conventional reliability learning cycle is inherently slow, and it is struggling to keep up with the presently accelerating pace of the PV industry. FPV represents a particularly challenging case, as the combination of diverse deployment locations (in terms of wave-heights, currents, salt- vs freshwater, etc) and diverse floater technologies (pontoon, platform, membrane, etc) gives a very large range of conditions to test, and limited transferability between cases. Therefore, there is a strong need to move to a more modelling-based learning cycle, enabling accelerated learning in the digital realm[9]. SuRE will develop modelling-based reliability assessment for three key stressors/degradation modes for FPV (Table 1.4). First, a modelling chain will be established capable of assessing mechanical loads throughout the FPV system and down to

specific components. This modelling chain can assess different combinations of technology, anchoring/mooring, and water body conditions. Second, the effects on the PV module of the increased humidity stress in the FPV configuration will be assessed, using gathered FPV humidity stress profiles as input. And third, the effects of increased levels of bird droppings and frequency and severity of hotspots will be evaluated with thermal modelling.

For the mechanical load modelling, the supporting structures used to keep the power producing and electrical components afloat and in place will be a central focus area. Assessing mechanical loads on these structures, especially when going to rougher sea-states, is a complex problem much beyond what is encountered for support structures in GPV. To address this properly, SuRE will take advantage of advances in floating wind technology. IFE's Wind department has, for more than a decade, developed an in-house aero-hydro-servo-elastic simulation software (3DFloat)[10] used for floating platforms for wind turbines, floating bridges and floating docks. 3DFloat is thus ideally suited for the FPV challenges at hand. It will be specifically tailored and adjusted for FPV in this project.

Pre-normative accelerated stress test development. By combining degradation mode statistics, quantified field stress levels, and insights from modelling, SuRE will develop an accelerated stress testing regime for FPV. By establishing the accelerated stress tests based on this triplet of input, SuRE will carve out a testing regime which will be easier to adapt and extrapolate to new waters and markets of FPV. This will be of vital importance to the industry as the market matures. The tests and their scientific basis will be disseminated in a white paper on reliability testing for FPV. Further, the developed accelerating stress tests will be used to assess the efficacy of reliability risk-mitigation strategies developed in the project in close collaboration with SuRE technology providers.

Technology specific design changes and materials selections for ensured and enhanced reliability. It is of outmost importance to ensure that all technology-specific innovations are compatible with reliability requirements. For CTI, the new design will lead to changes to the internal island forces, info on which is required to properly design the anchoring system, joints, mounting pieces for PV modules, and electrical components bringing the electricity to shore. Data and gathered experience from the SuRE pilot will be of central importance to this work. New material candidates with improved sustainability will be evaluated by accelerated stress testing. For ZIM, a central idea is to optimize the use of flexible joints in two cases: (1) between some (but not all) of the PV boats and (2) between the island and mooring lines. Each flexible joint is more expensive than a rigid connector, but the structural demands of the island, mooring, and anchoring system can be reduced with the right design. Lower forces and lower uncertainty in forces means an optimized design for conditions and therefore decreased steel use and anchoring costs. Lab-based testing will accompany component and plant design. The test parameters will come from a) measurements and correlation with metocean conditions at existing plants, and b) testing at the pilot site before construction. Measurements of conditions which accelerate ageing, and therefore feed into design needs, are forces, vibrations, motion, humidity, and time of wetness from splashing. With this information, relevant testing conditions will be set for dynamic loads on cable bending, inverter and transformer shaking, construction and connector bending, float and system buoyancy, mooring systems, and so forth. Component design updates include changing materials; improving protection for weathering, UV, and corrosion; and optimizing for efficiency and sustainability. The results become input to standard design practice for SSC 4 (moderate). For Sunlit, the floater design will be optimized to harvest potential reductions in CAPEX and LCOE. The model-chain for mechanical loads will be central to ensure that this optimization process is fully compatible with reliable operation in the targeted waters. The thermal exchange between module and water through the float will also be optimized; a lower operating temperature will have implications for most degradation processes as they are typically thermally activated.

Table 1.5 ZIM's approach to rougher waters in both generalizable WPs 1-3 and in WP 5

Materials and component design	• Corrosion protection for higher salinity and further corrosion categories • Accommodation of water icing in different scenarios • Long-time UV protection with HDPE additives • Structure design and material selection for profile design, connection, etc.
Modelling	• Higher wind speed and WMO SSC 4, including wave barrier concepts • Simulations for higher snow loads
Lab tests for individual components	• Corrosion tests to represent targeted new water conditions • Testing HDPE to represent targeted new water conditions • Fatigue simulations and tests for cables, inverters, connectors, etc. • Vibration of electrical components, connections points
Force tests	• Resilience of structure and components adapted from WP2 methods
Testing, 5 MW pilot	• Adapt system design to incorporate innovative components/concepts

1.2.1.3 Increased efficiency and cost-competitiveness of FPV power plants (WP3 + WP4-6)

The approach of SuRE to improve efficiency and cost-competitiveness of FPV will be two-fold. First, important knowledge gaps related to doing proper EYA of FPV will be closed. This will drive down cost both by improving bankability of FPV projects and by easing project planning. Further, the improved understanding will help the technology providers optimize their designs for improved efficiency and lifetime energy yield.

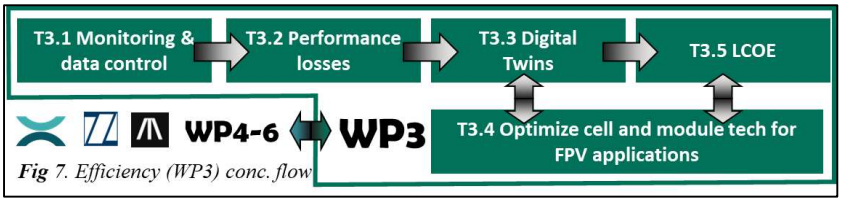


Fig 7. Efficiency (WP3) conc. flow

Analysis of FPV production data. Data analytics of FPV power plants is an active and relatively new research field, while more progress has been made towards advanced and robust data analytics for GPV. IFE, Fraunhofer and TNO have both published in scientific literature and worked directly with industry on PV power plant analytics. The research partners are international forerunners on this topic and have built substantial libraries of relevant algorithms developed both internally and externally. For determining performance ratio (PR), soiling, degradation rates, and other important parameters, we require data filtering and analytics algorithms optimized for the exact parameter in addition to well-instrumented sites. The research adheres to recommended standards for instrumentation, monitoring, and filtering. Because the existing standards and algorithms are based on experience from GPV only, a continuous part of SuRE is to assess to which degree these are sufficient. The analysis framework exploits renowned analytics libraries, particularly PVlib and Rdtools.

More accurate energy-yield assessments

Accurate PV module operating temperature in FPV applications. A frequently stated advantage of FPV over GPV is reduced module temperature from water cooling, resulting in a higher power conversion efficiency. Many generic and non-bankable claims have been made about enhanced cooling of FPV[11,12], and continue to be repeated even in high-impact reports[13]. These claims, however, under-communicate the strong dependence of module temperature on the specific FPV technology. Recent results and publications indicate that the cooling effect on typical pontoon-based floaters is, in fact, modest[11,12,14]. A clear understanding and quantification of cooling effects for different FPV floater technologies is currently lacking[15]. Uncertainty in predicting cooling-enhanced performance directly converts into EYA and LCOE uncertainties. Recently published best-practice guidelines from DNV GL state that it is "recommended to apply different thermal loss factors for different types of systems and different float technologies"[16]. Lacking appropriate simulation software and technology-specific recommended thermal loss factors, however, the current "best practice procedure is to determine the thermal loss factors for FPV systems through application of technology-specific measurements"[16]. To diversify available data for technology and climatic characteristics and address unanswered questions for performance and cooling of various FPV technologies, therefore, it is critical to collect and analyze field data from a range of systems. SuRE will accompany this analysis with modelling to establish climatic dependence of a) heat loss coefficients (U-values) for different FPV technologies, and b) influence of module temperatures variations over the power plant.

FPV and GPV mounting structures are very different. Standard GPV models to compute cell temperature and U-value, therefore, are not always valid for FPV. Detailed information on the applied materials and geometrical design of the different FPV systems will be used as a basis for the analyses. We will build complete thermal models for FPV power plants, including effects of conduction, radiation, air flow/convection and evaporation. Thermal and fluid mechanical modelling will be carried out with commercial computational fluid dynamics (CFD) software ANSYS FLUENT. The model will output PV module temperatures as a function of irradiation, air temperature, water temperature, and wind. The PV module temperatures will subsequently be used as input to calculations of performance, yield and, subsequently, LCOE of the FPV power plant in PVsyst.

Quantifying wave-induced mismatch losses for FPV. TNO introduced the WIML concept 2019[17]. IFE complemented the work with an alternative methodology in 2022[18]. Initial estimates have been computed for only a few wave states and no technology-specific calculations have been made. Computed losses are yet to be field-verified. DNV GL's best-practice handbook[16] suggests a simplified approach of using a ± 2° devitaion and setting the lowest performing value as the power output. Such coarseness lacks seasonal effects, wave state specifics and technology dependence: it reflects the lack of, and need for more, precise and standardized methodology to determine mismatch losses.

The effects of floater and module movements on energy output greatly depend on the system design. The number of PV modules per float, float dimensions, floater interconnection properties, electrical design, and choice of electrical

components are all important. SuRE will provide a classification for FPV systems, sea states and the potential effects of WIML. The goal is to establish WIML for different sea states to derive the net annual WIML for the three FPV technologies at any future site when sea states are known. We will build on exsisting models developed by TNO and IFE. To improve WIML prediction usefulness for different FPV technologies, hydrodynamic models will be introduced to define PV module movement as a function of sea state.

Hydrodynamic modelling capabilities will be tapped from the parallell SuRE developments to allow mechanical load modelling throughout the FPV systems. Subsequently, electrical models previously developed will be used to calculate the mismatch in each string. The effect of different inverters/maximum power point trackers will be analyzed; a detail not previously incorporated for WIML. Measures and cost efficiency of mitigating WIML, such as design of string size and layout, in addition to the effect of using optimizers, will subsequently be considered. The models will be validated in three steps. First, the orientation of individual floaters for different weather conditions, waves and sea states will be measured with accelerometers to verify the hydrodynamic models and to calculate in-plane irradiance for each PV module. Second, in-plane irradiation will be directly measured for each PV-module, using fast-response irradiation sensors to validate the choice of transposition models. Finally, the electrical model will be validated by comparing irradiance and corresponding production per module, and measured production per string. These steps allow calculation of the wave-induced mismatch corrected Performance Ratio. We suggest this concept to account for WIML in the same manner that temperature-corrected performance ratio is routinely used to separate inherent Performance of the Plant, as intended in Due Diligence contracts, from weather-induced variations.

Quantifying soiling loss rates and performance loss rates (PLR) for FPV. There are currently many myths surrounding soiling of FPV power plants. No reliable, openly available or published reference quantify this loss. In stark contrast, references for GPV soiling are substantial. Un-remediated soiling manifests in production data similarly as degradation and is indistinguishable from other degradation modes without the aid of additional sensors or dedicated algorithms designed to quantify soiling. The two losses, however, are fundamentally different because soiling losses are *recoverable losses* that can be mitigated with O&M routines, while degradation leads to a non-recoverable performance loss. Hard soiling, such as bird droppings, *can* also cause non-recoverable losses as it may generate hot spots with runaway module damage. This is however a secondary effect, and is avoidable by proper O&M measures. In SuRE, we aim to separate the recoverable soiling loss from the unrecoverable system PLR. Quantifying loss rates from each process provides valuable input to optimize cleaning routines and reduce OPEX for FPV power plants, as well as to improving energy-yield predictions.

Monitoring data from PV power plants can be used to determine soiling losses and (non-recoverable) PLR endogenously. Soiling losses are also commonly measured using soiling stations. Both methodologies hold expected challenges for FPV that SuRE will address. Within one FPV plant, spatial variations of soiling from bird droppings can occur. Whole-plant analysis using localized soiling stations is therefore inadequate. We will instead use a procedure developed by IFE to compute soiling loss rates from time series data[19]. The methodology was developed for utility scale GPV sites, with more homogeneous soiling and lower signal-to-noise ratio than what we expect for FPV. Several approaches will be applied to adapt the methodology to FPV, such as compensation for WIML, adaptation of data filtering, and using alternative performance metrics[20]. To verify the methodology and ensure that local soiling rates computed on inverter or string level correspond to soiling levels on site, infrared and visual image inspection will be performed for selected sites.

Choice of module technology for FPV applications. Economy of scale and little knowledge of FPV specific stressors are reasons why there are small or no differences between PV modules used for GPV and FPV applications. With a rapidly growing FPV market and an increasing knowledge base, this may change. MetSolar has a PV module production line and an in-house R&D department. The starting point to identify potential FPV specific design adaptations will be to analyze production costs for different cell and module technologies and their resulting field efficiency. Field efficiency will depend on the mounting technology and application, which will be calculated for the three FPV technologies in the consortium and deployment in Europe. For FPV, where the mounting structure costs are higher compared to GPV, the tradeoff of moving to technology with higher power-conversion efficiency for higher cost is an important consideration, taking overall LCOE into account. The choice of adaptations to MetSolar's production line will follow from the analysis, but three likely pathways have already been identified. The first possible pathway is related to the mounting angle in FPV applications, which is usually lower than optimal due to the reliability and cost of the BoS structures. Standard PV module designs include coatings on the module glass and the silicon PV cells to transmit as much light as possible into the active parts; these designs are dependent on the angle of incidence. It is possible to increase transmission of light at non-standard angles into standard PV cells by

modifying the module glass. One such option is prismatic glass, with small physical modulations that aid light transmission, and thereby electricity production, without needing to modify anything other than the glass that encapsulates the module. A second path is a change towards higher efficiency cells, which also infers increasing the cell size to M10/M12. The higher BOS cost of FPV might make this a particularly interesting option. The third potential pathway is to analyze the cost-efficiency and implement a selection of the mitigation measures for PV modules in FPV applications, identified in the work described in Section 1.2.1.2.

LCOE calculations. To (1) compare different technological solutions and (2) assess the economic benefits of different technological advances made within SuRE, we must establish a framework for which to perform LCOE calculations. The framework will be directly linked with the knowledge obtained through the development of more accurate EYA, to consider the economic effects from increased efficiency for different water bodies and locations. Further, it will adapt enhanced uncertainty propagation methods for LCOE calculations, going beyond the classical P50-P90 schemes. The accuracy of these standard schemes relies on assuming small standard deviations of the input variables, which need not be the case and which the enhanced approach does not presuppose. As a result, decision making on FPV plants is facilitated by (1) providing reliable LCOE averages *and* reliable uncertainties, and (2) computing probability distributions of net-present value and thus the total probability of profitable investment.

Technology specific design changes and materials selections for enhanced efficiency. All three technologies will draw benefits from the improvements in 1) EYA-estimates and LCOE calculations, 2) general O&M developments such as tools to optimize cleaning cycles, 3) thermal modelling to optimize the system designs for maximized cooling, and 4) improvements in estimations of WIML to guide necessary adjustments in design to maximize power output, especially when moving to higher SSCs. For CTI, a key feature of their innovative design is an increased surface efficiency by reducing maintenance walkways. To do so properly, however, requires maintenance methods which are precisely matched to the design, which will be tested and developed in SuRE.

1.2.1 Research and Innovation activities linked to SuRE

The partners in SuRE are experienced in R&D projects, from Horizon 2020, Horizon Europe, national projects, and bilateral industry projects. The R&I carried out in SuRE builds on previous and ongoing projects. The consortium details 28 FPV related R&I projects in Part A. Only the R&D activities with the most direct links to SuRE are shown below. EU projects that SuRE will link to are given in grey background.

Table 1.6 Selected research and innovation activities whose results feed into SuRE

Project	*Project aims and outputs used in SuRE*
IEA PVPS Task 12 and 13 (Global) *Fraunhofer, IFE, TNO*	In Task 13 Fraunhofer is Leading Subtask 2 (ST.2) on Performance and Reliability of PV. Within ST2, the activity on FPV is lead by IFE with TNO and 15 research/ industry participants. Workshops, webinars, publication of FPV report in 2025. In Task 12, TNO is active on PV Sustainability, TNO is active on LCA.
Standardization TC82/2106 *Fraunhofer, BayWa*	Standardization effort within the International Electrotechnical Commission (IEC) Technical Specification, *FPV power plants – Design guidelines and recommendations*. Well-coordinated with SuRE through Fraunhofer and BayWa.
SUREWAVE (HE 101083342) *Sunlit Sea*	Securing optimal behavior of aero & hydrodynamic and structural integrity for offshore FPV. Concepts and results of the connectors, whole FPV system, electrial output, and cost-efficiency will feed in to WPs 2-6.
Trust PV (HE 952957) BayWa	Performance and Reliability of PV plants, with a focus on GPV, but with a smaller activity on FPV. Coordinated with SuRE through BayWa's participation in both project and IFEs contact with EURAC (coordinator of Trust PV).
ZWIMP & Andijk (Dutch funding) *Deltares*	Continuous monitoring data below FPV system and on a reference site. 3D Modelling of FPV effects on water temperature. Both data and modelling approach for FPV will be input information to WPs 1 and 4-6.
Serendi-PV (H2020 953016) *Fraunhofer*	Continued reduction of LCoE for PV and to make it possible to integrate a rapidly increasing share of PV power into the power networks. Yield simulations for FPV concepts used in WPs 3 and 4-6
Pilot Oostvoornse Meer (Dutch TEHE116900) *TNO*	FPV pilot on a near shore lake with brackish water from 2019 until 2023 with three designs. The electrical, mechanical properties and water quality are continuously monitored and analysed. Methods and models used in WP 1-3, 4-6.
AC Yield (Dutch TKITOE	Research project on thermal behaviour of a 2 MWp FPV plant to gain insight on

1821301) *TNO, BayWa*	the cooling effects of FPV vs GPV plant. Cooling models as used in WPs 3, 4-5.
MariSol *(Norway 317808) IFE, Sunlit*	Development of Sunlit's FPV concept. Work on reliability and performance analysis will be directly fed into WPs 2, 6.
ECLIPSE (MSCA 101065785) *2022-2025*	Ecosystem impacts from FPV. Methods, results, and assessments from ECLIPSE and SuRE can be shared, compared, and tested for broadest applicability because of different specific waterbodies of application.
BOOST (H2020 965671)	Project on offshore FPV, coordinated by Fred Olsen Renewables and with IFEs long-term industry-partner Ocean Sun as one of the main FPV technologies to be developed. Well coordinated with efforts in SuRE through IFEs close contact with Ocean Sun and Fred Olsen Renewables.
NATURSEA-PV *(HE 101084348) 2022-2026*	Developing biobased, antifouling structures for FPV. Materials, results, and environmental impact from both projects can be shared for accelerated materials development in both inland and offshore FPV applications.

1.2.2 Interdisciplinary work

Development of sustainable FPV power plants with long lifetimes and circular use of materials requires cooperation between a large set of disciplines, including, but not limited to, power plant design, hydrodynamics, statistical analysis, biology and biodiversity, sustainable plastics, aluminium and steel manufacture, PV modules, business case development, exploitation of research and technical outcomes, technoeconomic analysis, and public engagement through visual arts. The 12 partners of the SuRE consortium bring a balance to achieve the Outcomes set in the Call and to contribute beyond the state of art. Through the art-meets-science vision of Compáz who leads the communication activities, SuRE promotes public acceptance and mutual understanding at the society-policymakers-developer nexus. Such interactions are key to rapid, responsible, and sustainable renewable energy deployment.

1.2.3 Gender dimensions in the Research and Innovation content of SuRE

The SuRE consortium recognizes the importance and potential of fostering gender equality, in line with the strategy on gender equality adopted by Horizon Europe: A Union of Equality: Gender Equality Strategy 2020-2025. Gender analysis will be implemented in the continuous workflow during the SuRE project. We evaluate past innovation practices when we design new components (for instance the trend of moving to larger PV modules, making it difficult for women installers), and ensuring women participants in the various teams to broaden perspectives and addressing sex and gender when analyzing users and markets. We have identified areas in SuRE where gender is relevant to enable us to implement gender dimensions in the initial planning of the R&I, basing the analysis on methods in *"Gendered Innovations: How Gender Analysis Contributes to Research".* We highlight four areas in SuRE where gender dimensions clearly impact the planning of our developments:

- ✓ Development of technical solutions such as design of walkways and O&M routines for FPV power plants.
- ✓ Development of visual tools or illustrations, for communication/dissemination purposes, user manuals, or O&M software, the consortium will take care to develop gender-neutral examples, images and visualization.
- ✓ Communication and dissemination activities, including work related to public acceptance of FPV power plants and policies related to permitting. The relationship between gender and environmental impact, and gender and technology are relevant here. Compáz addresses this part of SuRE specifically (See WP7 and Sec 2.2.1)
- ✓ Training of researchers and personnel for the renewable energy industry is an important Societal Impact in SuRE.

It is of outmost importance for the gender balance within the energy industry to focus on the gender dimension in recruiting and training. A recent report from IEA[21] shows that despite making up 39% of global labor force – women only account for 22% of the traditional energy sector. For management levels the numbers are even lower[22]. The low employment rate of women in the energy sector is also certainly in part due to a pipeline problem, since there is still a disproportionately low number of women with STEM degrees, reducing the pool of potential female applicants for some firms and occupations. If women are not advancing in the energy sector, there will be fewer women role models and mentors to attract more women. The barriers women face in the energy sector are similar to those they face elsewhere in the economy. However, the challenges of the energy sector are more pressing since the sector is going through a process of transformation; the clean energy transition will require innovative solutions and business models to be adopted and greater participation from a diverse talent pool. The training of personnel, and women in particular, is important for the consortium. The aim of both the research in SuRE, and the long-term goal of the participating companies, is to develop and deploy new, RE technology. It is a benefit for the innovation and sustainability of the individual companies to encourage a larger fraction of female employees. Increased visibility of women and non-

binary scientists can inspire members of underrepresented communities to join the field, hence we will strive for gender balance in all dissemination activities. Both SuRE and its communication partner, Compáz, are led by women. Towards the end of the project period, we will perform an evaluation amongst the partners on how gender perspectives were implemented in practice, what consequences it had for the research and development in the project, and whether the work to address gender dimensions in SuRE also influenced the company in a broader sense, which will be included in the Final Report.

1.2.4 Open Science Practice

SuRE will adopt the Horizon Europe Regulations and the Model Grant Agreement, developing a clear set of actions to ensure that the outcomes are fully compliant with the stablished Open Science rules. The partners will be responsible for publishing non-confidential outcomes in local and international press, at conferences in the form of papers, and in peer-reviewed scientific journals. SuRE will ensure that all the publications can be read online, are downloadable and printable and freely available online with no restrictions on use, by depositing them in a repository. All the peer reviewed articles will be made Open Access after publication, without embargo periods, following the rules set by the European Commission. Several open access repositories will be considered, such as the Open Access Infrastructure for Research in Europe (OpenAIRE), the Registry of Open Access Repositories (ROAR) and the Directory of Open Access Repositories (OpenDOAR). A decision will be taken for every specific case and all the partners engaged in publication activities have a dedicated budget to cover the open access fees. The Consortium fully adheres to the principle “as open as possible, as closed as necessary” set by the European Commission to allow the outcomes to reach society whilst protecting the IPRs of the companies involved in the co-development of the technical solutions for the RIs. Most WPs will have several public deliverables so that the knowledge can be shared freely with all relevant stakeholders (research community, industry, and regulators). Deliverables that are considered confidential will be shared with stakeholders (if relevant) after signing an NDA. Public research data generated will be made openly accessible via the online repository ZENODO, as soon as the research has been published. ZENODO is a free of charge repository developed by CERN within the EU OpenAIRE project to support Open Data management for EC funded multi-disciplinary research. In general, the SuRE consortium focuses on ensuring open access to research data. However, because of the sensitive nature of some data parts, in certain cases, these parts may remain confidential and not be accessible to third parties.

1.2.5 Research data management and management of other research outputs

All SuRE data storage will be in line with General Data Protection Regulation 2016/679 and The Consortium Agreement will cover all aspects of data and data management. Individual partners will curate their data, as overseen by the coordinator. Partners decide whether data storage is local or in the cloud. Green archiving for confidential information will be used for commercially and/or technologically sensitive information, with measures for eventual access after protection or declassification. SuRE will develop a project data management plan to handle data for exploitation and (in the case of non-confidential data) dissemination. The data management plan, deliverable D8.2, will outline principles and processes for data collection, organisation, management, storage, security, analysis and sharing. A first draft will be ready by M6 and will be updated periodically during the project. Data storage in appropriate repositories (e.g., Zenodo) will ensure data viability after the project completion. Restrictions to data access will comply with the IPR standards set by the consortium as well as Horizon Europe principles on open access research (as open as possible and as closed as necessary). Ensuring competitiveness and enhancing market potential of the products and processes developed in SuRE may require restrictions to data. IFE will oversee the process to guarantee that both dissemination and exploitation comply with the conditions defined in the Grant Agreement.

Do no significant harm

SuRE methodology complies with Article 17 of EU Regulation No 2020/852 (“do no significant harm”) because we treat advances through co-optimization of economic and life-cycle environmental parameters; improved water quality and biodiversity; decreased material use; and increased recyclability. This working lens ensures coordinated positive movements towards all 6 environmental objectives: Climate change mitigation; Climate change adaptation; The sustainable use and protection of water and marine resources; The transition to a circular economy; Pollution prevention and control; and The protection and restoration of biodiversity and ecosystems.

#§CON-MET-CM§# #§COM-PLE-CP§# #§REL-EVA-RE§#

2 Impact #@IMP-ACT-IA@#

In this Chapter, we will show which Outcomes and Impacts the actions in SuRE will lead to. We will also detail the plans for the dissemination, exploitation and communication of the results which will enable the realization of these Outcomes and Impacts. A short summary of the most important long-term Impacts from SuRE follows below.

1. ***Reduced cost and improved efficiency of sustainable RE and their associated value chains.*** The deployment of FPV in Europe will accelerate, as investors and project developers will have a better understanding of risk mitigation and electricity yield forecast, and an improved business case through improved FPV designs and reduced LCOE.
2. ***Increase knowledge on the environmental impacts, and enhance sustainability, of RE technologies along their lifetime and value chains.*** The further deployment of FPV in Europe will be implemented with a substantial higher level of circularity and a substantial lower risk for the environment.
3. ***Input to policy making generate enhanced security and autonomy of energy supply in EU, while accelerating the green transition.*** The potential area available for FPV deployment in Europe will substantially increase as permitting authorities better understand the conditions when they can and cannot issue permits. The competitive advantages of the European FPV value chain are strengthened, reducing dependency on imported components.

2.1 Project's pathways towards impact

The consortium has identified 5 important Target Groups to maximize the impact of the results from SuRE. More information on each of these groups are described in the Table below. The Target Groups are used to identify who will benefit from the Outcomes and Impacts (in Section 2.1), and who SuRE needs to communicate with to achieve the Outcomes and Impacts (in Section 2.3).

TARGET GROUP	SEGMENTS		FOCUS
1. Main end-users	• Utility companies • Investors	• Consultancies • Engineering companies	HIGH
Includes main off takers of the FPV technology developed within the project. SuRE engages with this target group to i) Ensure that the developments reach a large market beyond the off takers within the consortium, ii) Contribute to expanding the FPV market, iii) Reduce investor hesitancy from uncertainty			
2. FPV industry	• FPV technology providers • Asset managers • Certification bodies	• Materials & Component suppliers (incl. software)	HIGH
Includes the premise suppliers and the broader FPV industry. SuRE engages with the wider FPV industry to i) Ensure that whole FPV industry can make use of established best-practices and openly available results from SuRE, ii) Increase the availability of materials and components with the required quality and sustainability, important for scale-ups and further improvements in sustainability and cost-efficiency along the value chain.			
3. Public authorities	• Policy makers • Concession granting bodies	• European Commission • Regulators	HIGH
Includes entities responsible for setting rules and incentives for designing and implementing FPV power plants, and strengthening Europe's energy autonomy, human capital, and competitiveness. SuRE engages with public authorities to improve the knowledge base needed for regulations and legislation for FPV deployment. Access to scientific knowledge and best-practices is key to ensure sustainable, safe, and cost-efficient deployment of FPV.			
4. Scientific Com.	• Energy and power systems • Environmental Sciences	• Marine technology • Materials technology	MEDIUM
FPV is highly interdisciplinary in nature, and SuRE will interact with a broad scientific community to; i) ensure that the open results from SuRE can and will be utilized to further deepen the understanding of various aspects of FPV and enable further improvements in FPV technology and deployment practices, ii) ensure that the consortium has access to the relevant, new and updated knowledge from within the field of FPV and from other disciplines.			
5. General Public	• General public		MEDIUM
SuRE encourages a common understanding and trust in the positive impact of increased production of RE technology, and deployment of RE in Europe. Highly accessible content is produced to engage with the society.			

2.1.1 Contribution of SuRE towards the Impact specified in the Destination and Outcomes of the Topic

SuRE will contribute substantially and uniquely to all expected outcomes of the call, and to five of the Impacts of the wider destination theme "Global leadership in RE".

Impact 1: Reduced cost and improved efficiency of sustainable RE and their value chains

Outcome 1: New FPV designs that reduce CAPEX and OPEX, maximize energy output and reduce LCOE.

In SuRE three leading European FPV suppliers will develop and verify new designs of different components of their FPV technologies. Choices and changes in component design will be guided by new predictive yield models with higher accuracy than what is currently available. Tabular forms of these results will be presented, enabling the gained knowledge to be beneficial also for deployment of other FPV technologies. The calculations will be performed for different scales and sea states. The models will also be used to optimize maintenance routines and reduce costs of O&M for FPV power plants. The advanced module and system concepts developed by CTI and ZIM will be demonstrated in FPV power plants > 5 MW**.** The results from SuRE will also include quantification of stressors and suggested alterations to accelerated stress tests for GPV modules according to the changes in experienced stress. The direct outcome of this work by the end of the project period is a significant improvement in the designs of the three FPV technologies from CTI, ZIM, and Sunlit, reducing both their CAPEX and OPEX, while optimizing the energy output. In sum this leads to a significant reduction in LCOE. Specific goals for the cost reductions are given as KPIs. In the medium-term, the LCOE reduction will be most prominent for the three FPV technology value chains represented in the consortium, but many of the results will be transferable to other FPV technologies and will in the longer term contribute to a reduced cost and improved efficiency also for other FPV value chains. The potential impact in the longer-term is that FPV will be more cost-competitive compared to fossil fuels and land-consuming RE technologies, which can significantly increase the deployment rate.

The results are linked to **HLO 1** in this proposal and will be accomplished through the work in **WP2, 3, 4, 5 and 6**. The following **KPIs** will be efficient measures of the achievement of these goals: **1)** Deviations between modelled and measured production data for the monitored sites is reduced, indicating increased accuracy in predictive yield models, **2)** A 10% reduction in LCOE for the new CTI technology, **3)** A reduction of 15% in BoS costs for Zimmermann's FPV technology, **4)** A reduction in LCOE of 60% for Sunlit's FPV technology **5)** One new, accelerated stress test for FPV modules in a near shore environment suggested. **Target groups:** 1) main end-users, 2) FPV industry, 3) Public authorities, 4) Scientific community, and 5) General Public.

Impact 2: Enhanced security and autonomy of energy supply in EU, while accelerating the green transition

Outcome 2: Expand the application of FPV for inland and offshore waters.

Through SuRE, ZIM will expand their FPV design standards from SSC 3 (WMO SSC: *Slight*, wave height 0.5 – 1.25 m) to SSC 4 (*Moderate*, wave height 1.25 to 2.5) through optimization and increased structural integrity of their steel components, and development of new interconnects and mooring solutions. Sunlit has built pilots up to SSC3/4 and aim to validate their design up to SSC 5. The design modifications of all three FPV designs will enable them to expand the application of their technologies to water bodies with higher SSCs and higher salinity.

The results are linked to **HLO 2** in this proposal and will be accomplished through the work in **WP2, 4, 5 and 6**. **KPIs: 1)** New design standard for ZIM Float for SSC 4. **2)** Validation of Sunlit technology for SSC 5. **3)** New accelerated stress tests specifically designed for FPV developed addressing three key stressors. **4)** Publication and revision of pre-normative procedures. Adoption of procedures by FPV producers and installers. **Target groups**: 1) Main end-users, 2) FPV industry, 3) Public authorities, 4) Scientific community, and 5) General Public.

Impact 3: Enhanced sustainability of renewable energy value chains, taking fully into account circular economy, social, economic and environmental aspects in line with European Green Deal priorities.

In SuRE, enhanced sustainability of RE value chains will be adressed primarily through reduced carbon footprints of FPV power plants due to reduced material consumption (reductions in HDPE, Steel, and Alumium usage, respectively), use of more recycled materials (post-industrial HDPE to post-consumer HDPE or PP, as well as recycled aluminum), and focus on deployment that has a positive impact on the local environment.

The results are linked to **HLO3** in this proposal and will be accomplished through the work in **WP1, 3, 4, 5 and 6**. **KPIs: 1)** Reduced carbon footprint for ZIM Float by 20% **2)** Documented, reduced carbon footprint for CTI technology **3)** 50% reduction in Al consumption for Sunlit, models for using recycled Al. **4)** The KPIs for Impact 1 and Outcome 1 are also relevant for Impact 3. The **Target groups:** 1) Main end-users, 2) FPV industry, 3) Public authorities, 4) Scientific community, and 5) General Public.

Impact 4: More effective market uptake of RE to support commercialization & provide input to policy making.

SuRE aims to reduce the observed hesitance to enter and/or invest in FPV and to increase societal acceptance of FPV. The consortium has identified uncertainty as a hindrance for greater market uptake of FPV. There is uncertainty related to sustainability, lifetime, and yield for FPV power plants. SuRE address all these three aspects in the generalized WPs with the aim of making more knowledge publicly available for industry stakeholders. SuRE, through substantial effort in WP7, also has a strong commitment to communicate with European citizens and policy makers. The results related to sustainability are considered to be of special relevance for a broad audience.
The results are linked to **HLO4** in this proposal and will be accomplished through the work in **WP1, 2, 3, and 7**. **KPIs: 1)** Facilitate communication between public and FPV providers through artistic platforms **2)** Streamline LCOE calculations to accelerate investment process. **3)** Paper detailing inputs of EYA for FPV in PVsyst. **4)** Publication of sensitivity of FPV plant decisions on LCOE. **Target groups**: 1) Main end-users, 2) FPV industry, 3) Public authorities, 4) Scientific community, and 5) GP.

Impact 5: Increase knowledge on the environmental impacts of RE techs along their lifetime and value chains.

Outcome 3: Minimize the environmental impact of FPV technology for inland and offshore waters.

A result from SuRE will be a set of guidelines including relevant parameters, methods for monitoring, and reference values, to assess the direct impact of FPV on aquatic systems and biodiversity. The impact of FPV on ecosystem biodiversity will be quantified, and optimizations and mitigation measures will be investigated to further reduce negative impact and enhance positive impact on the environment. These results will enable policy makers and concession granting bodies a solid scientific foundation for regulating FPV deployment, do proper and efficient permitting, and expand application of FPV while minimizing any negative environmental impact and maximizing the potential positive impacts on local environment and biodiversity.
These results from SuRE are linked primarily with **HLO3** and will be accomplished through the work in **WP1, 4, 5, and 6**. **KPIs: 1)** Protocol for measuring hydrological and ambient variables to model the interaction of FPV with waterbodies, **2)** Published results on biodiversity mitigation measures, and on effect of coverage ratios for European waterbody types, **3)** Internally agreed pre-normative procedures for assessing FPV sustainability form basis of published guidelines for sustainability monitoring of FPV power plants. **Target groups:** 1) Main end-users, 2) FPV industry, 3) Public authorities, 4) Scientific community, and 5) GP.

2.1.2 The Key Impact Pathways

■ *Scientific Impact* ■	***Creating high-quality knowledge***

In SURE the **four leading research institutes** on FPV in Europe is collaborating with the leading FPV technology providers to push the forefront of FPV science and technology. FPV is a relatively new technology, and access to high quality scientific literature based on field data and monitoring is very scarce. There is a great need for more accessible, openly available and high-quality scientific, and popular scientific, literature. In SuRE the research partners will analyse production and monitoring data from a number of commercial FPV sites. We expect that much of this information can and will be shared in peer reviewed publications and will have a huge impact on the knowledge base for FPV. The publishing work will predominantly be carried out in WP1, 2 and 3. List of papers with preliminary titles is provided in the Dissemination Plan in Section 2.2.

Indicator:Number of peer-reviewed papers & conference proc.	9 peer-reviewed + 4 conference proceedings

Strengthening human capital in R&I

With the tremendous upscaling of RE deployment, there is a lack of skilled personell in the business. Technical careers and education in solar jobs must be promoted as a matter of priority to solve the sector's issue of lacking skilled labor. In SuRE, one PhD student (Deltares/Utrecht University) is fully funded by the project, 4 PhD students are associated with the project and 12 Master's students will be educated (University of Freiburg, University of Oslo and Utrecth University). Fraunhofer regularly offers workshops and training for the FPV industry. This will be an excellent platform to disseminate the results from FPV to ensure uptake of updated knowledge. Through the project, the topic of FPV will be introduced at the University of Oslo (UiO) through the courses TEK5330 and TEK9330. The course is taught by personell from IFE and is currently covering general aspects of PV systems, including BIPV, but not yet FPV. A 3-hour lecture on FPV will be developed and included in the course. The course reaches 15-20 students per year. The lecture material will be developed; the lectures will continue after the project's end. Fraunhofer will introduce FPV to the Solar Energy module in the Sustainable Systems Engineering Course at University of Freiburg. Lector Miguel Dionsio Pires from AERES University of Applied Science is part of the Deltares-SuRE team. In his course 'Innovations in monitoring water quality' at AERES he will disseminate the results of SuRe in his yearly courses on water quality. Through SuRE he will

engage several students in fieldwork and research projects on FPV and 20 students a year at AERES will be educated on the environmental impact of FPV. SuRE will host an annual FPV summer school for 20 students. Funds are also allocated in SuRE to enable researcher upskilling and mobility among research partners and industry for visits. This will contribute to broadening the knowledge of the researchers on exchange, and build strong, long-term interaction between the partners in the consortium. The monitored large scale pilots and utility sites give the researchers partners access to unique infrastructure.	
Indicator: Number of researchers having benefitted from upskilling activities in project (through training, mobility, and access to infrastructures)	1/4 full/associated PhD students, 12 masterstudents + 150 students (over 5 years), 20 students on summer school, access to unique infrastructure ~ 20 researchers, mobility ~ 2
Fostering diffusion of knowledge and Open Science	
In several of the market segments covered by the SuRE projects, companies in direct competion (CTI/ZIM/Sunlit, BayWa/LKY) collaborates in the belief that collaboration foster mutual benefits; for the individual companies and for society through safer and more rapid deployment of FPV. The companies have found common grounds to collaborate on generic topics relevant for the whole FPV industry; sustainable deployment and a sustainable value chain, development of reliability test procedures and standards for FPV components and development of accurate models and tools for EYA of FPV power plants.	
Indicator: Share of Funded Project (FP) research outputs (open data/ publication/ software etc.) shared through open knowledge infrastructures (open access)	All publications in open access, recommondations for commercial softwares for FPV and environmental guidelines
■ *Societal Impact* ■	***EU policy priorities and global challenges through SDGs and R&I***
SuRE is strategically oriented to adress EU policy priorities and global challenges. Below is a summary of how SuRE will contribute to goals and indicators in central policy documents and SDGs.	
	**7.1** and **7.2.** SuRE provides new avenues for RE, whose source is cost-competitive and environmentally beneficial. **7.b.** Near-shore FPV is an ideal electricity source for small island developing States.
	**8.2**. The core of SuRE is to bring out the potential of FPV as a sustainable, renewable, and labor-intensive energy source through technological upgrading and innovation.
	**12.5.** Through the materials suppliers and technology providers, developments in SuRE are tied to recyclability, reduced waste, and end-of-life materials recovery.
Green Deal, RePower EU, EU Solar Energy Strategy: As PV implementation expands, issues related to land access are increasing, and FPV is a promising solution to reach the RePower EU targets for PV in a sustainable manner, without sacrificing local environmental concerns or conflict with other use of land. A main outcome for SuRE is to increase the deployment rate of FPV, in Europe and gloablly.	
Indicator: Number and share of **outputs** aimed at addressing specific EU policy priorities (including meeting SDGs)	100%: All the specific objectives of SuRE, and each of the 11 innovations (Table 1.1.3), contribute to specific EU policy priorites.
Strengthening the uptake of research and innovation in society	
Our approach to strengthen the market uptake is to reduce uncertainties and thereby reduce hesitancy of investing in FPV power plants, and to contribute to disseminate knowledge to a broad audience to reduce conflict, and thereby accelerate FPV. Reduced uncertainty includes improved accuracy of EYA and streamlined LCOE calculations. To reduce conflict, we seek to foster multilateral dialogue among residents, permitting bodies, research teams, FPV industry. WP7 Communication and Dissemination integrates the FPV piloting processes with public engagement through art. The WP leader, Compáz, combines their vision for both art and PV to maximize uptake of new technology by ensuring FPV suits both producers and consumers.	
Indicator: Number and share of Funded Projects where EU citizens and end-users contribute to the co-creation of R&I content.	Citizens and endu-users contribute to the co-creation of R&I in SuRE. The relevant tasks will be carried out in WP3 and WP7
■ *Technological/economic Impact* ■	***Generating innovation-based growth***

The SuRE consortium provides a uniqe arena for innovation-based growth in the European FPV industry. Projects where firms are attempting to forge new ties in new collaborative R&D efforts and make R&D strategies that are more accessible to other researchers and firms, are more likely to successfully diffuse and exploit the knowledge generated by the project. Projects with such attributes are more likely to yield economic benefits, all other things being equal, that extend beyond the firm directly engaged in the project. In SuRE the leading companies within FPV in Europe are working together to enforce the world leading competence and position of the European FPV industry and increase the deployment rate of European FPV technology, in Europe and globally. Through SuRE a number of innovative products, processes and methods will be developed, exploited by the partners in the consortium and efficienty communicated to the wide FPV industry, governance and public. The list of new producs, processes and methods developed in SuRE is provided in Table 1.1.3.

Indicator: Number of innovative products, processes, or methods from FP & Intellectual Property Rights applications	5 new products, 2 new processes, 4 new methods. See Table 1.1.3 for details.

Creating more and better jobs

New solar power generation capacities are highly cost-competitive, but yet the most job-intensive among low carbon and RE technologies. PV creates about 6 jobs per MW of capacity in manufacturing, construction, and installation and 1.2 - 4.8 in operating and maintenance[23], which means 7 to 11 FTEs/MW deployed. As a significant fraction will be employed within production of cells and modules, we will assume that only 10% of the FTEs/MW is employed within the part of the value chain covered by BayWa/ZIM and CTI/LKY, i.e., 1 FTE/MW. Sunlit covers a smaller and less labor-intensive part of the value chain than CTI/LKY and ZIM/BayWa as they do not deploy the systems. Market predictions for FPV vary greatly, also indicating the sensitivity to cost predictions for FPV, but an increase of **25% annually** is in the middle of the span of the predictions, and the needed annual growth rate of PV if we are to meet zero emission in 2050. However, these predictions are highly uncertain. *The combined results of LCOE reductions, increased application offshore, and reduced uncertainty related to permitting and sustainability-concerns, all outcomes of SuRE, will lead to a more rapid market growth than the benchmark scenario of 25%.* ZIM/BayWa's SuRE-target is to reduce BoS cost with 15%, translating into a ~ 5% reduction in LCOE. Combining the LCOE reductions with the plans for enhanced sustainability, dissemination and exploitation, CTI estimates that their developments in SuRE will increase their market share to 20% by 2028. Sunlit plans to have an establish production-line of 11 MW by mid-2024, and an upscaling to 455 MW by the end of 2026. *FTEs in production of other components and related value chains not included in these estimates. The estimates only look at additional MW and FTEs beyond the benchmark 25% market growth.*

	Current market share	Target ⇩ LCOE	Target market share	Annual growth	Additional Installed 2026	Additional Installed 2030 (MW)	Additional FTEs 2026	Additional FTEs 2030
ZIM	7%	⇩ 5%	12% (2030)	~ 40%	40	300	~ 40	~ 300
CTI	15%	⇩ 10%	20% (2028)	~ 35%	150	300	~ 150	~ 300
Sunlit	NA	⇩ 60%	7,5% (2026)	~ 75%	445	550	~ 50	~ 60

Indicator: Number of FTE jobs created, and jobs maintained in beneficiary entities for the FP project.	FTEs working in SuRE: ~20 FTEs/year. Additional FTEs within the consortium: by 2026 - 240, by 2030 – 660.

Leveraging investment in research and innovation

- To fully exploit the developments carried out in SuRE CTI must invest an estimated 2M€ in new production tools to produce and assemble the targeted 600 MW capacity.
- Two 5 MW pilots will be installed during SuRE. The external investments related to the pilots will surpass 5 M€ each.
- With the aid of the cost reductions in SuRE, Sunlit plans to increase the production to 455 MW/year. This goal will require a 23 M€ investment in production facilities and automation. A 105-kW pilot will be realized during the project period, with a value of ~ 150 k€. Financed outside of SuRE.

Indicator: Amount of public & private investment mobilized to exploit and scale-up the SuRE results	Production lines: 2 + 23 M€ = 25 M€. Pilots: 12 M€, SuRE leverage=1,2 M€. **Total = 38 M€**

2.1.3 Requirements and potential barriers to outcomes and impact

Table 2.1 lists potential barriers arising from factors beyond the scope and duration of the SuRE that may influence whether we are able to achieve the planned outcomes and impacts.

Table 2.1 Potential Barriers

Potential Barriers		How SuRE will address the barriers
Risk of technology	Technology development is a high-risk activity, and the success rate can be low.	Close collaboration between industrial actors and research organizations along all stages of development facilitates efficient innovation. Tight follow-up, support and help to develop realistic progress plans for the developers are crucial, with particular emphasis at the decision gates.
	Industrial secrecy may impact results and uptake beyond the project.	SuRE will be governed by a proper partnership agreement, which will smooth this barrier and will outline the necessary fine balance between conflicting interests.
Policy	Pre-mature policy decisions leading to FPV regulations based on precaution instead of scientific data	Currently, there is little high-quality, quantitative research available to base policies on, and many of the models published has not been verified with field data. Through SuRE, the leading international research institutes within FPV will access unprecedented amounts of field data from utility scale FPV sites. SuRE will make quantitative knowledge and verified models available as a basis for new policies. This is a prerequisite, but no guarantee for sound regulations. SuRE plan strategic dialogue with the relevant stakeholders to improve the dissemination, understanding and uptake of the research results.
	Negative media attention resulting in increased public opposition towards FPV power plants	Massive deployment of RE can easily lead to conflicting opinions on land use. Our guidelines on sustainability will be of high value to reduce the risk of unsustainable deployment and public opposition. Our dissemination and communication plan will be the main tool both to share the results and facilitate sustainable installations, but also to communicate information in an accessible manner to a wider public.
Public acceptance	LCOE of competitive technologies is reduced more rapidly than FPV resulting in lower market shares	The main strategy of SuRE is to contribute to fulfill the potential for cost reductions within FPV. PV is an industry with extreme cost reductions and learning curves. FPV is a relatively new application within PV, and the consortium evaluates the potential for cost reductions to be large. Even with modest cost reductions, FPV have competitive benefits in market with land scarcity and a great potential for shared infrastructure.
Market	The future market for FPV is smaller than foreseen	SuRE address what we consider the two main concerns for the continuous growth of FPV deployment; cost and sustainability & permitting.

2.1.4 Scale and significance of the project's contribution to outcomes and impacts

The International Energy Agency (IEA) reported that for the world to reach the goal of net zero emissions by 2050 there will have to be an annual average solar energy generation growth of 25%[1]. In 2020, solar generation increased 22%, resulting in the IEA categorizing solar photovoltaics (PV) as 'more effort needed' [24]. FPV is a clear addition to the PV industry, adding to overall growth while avoiding increasing competition with land use. To do FPV correctly, however, requires the broad view of SuRE. ***The SuRE Consortium is capable of making a remarkable impact on the LCOE of FPV power plants.*** Frequently hailed as the third pillar of the global PV market, FPV are gaining traction around the world The total installed capacity has surpassed 3 GW and the annual installed capacity of FPV is currently around 1 GW. The cost of FPV varies substantially, depending on market and local conditions, but assuming a cost of 1.5 €/W, the current value of the annual FPV market is around €1.5 billion. However, the variation in the estimates of growth and future market shares for FPV is striking. Published estimates of the annual CAGR varies from 5% to 43%[25]. If we assume a value of €1.5 billion in 2021 and a 25% linear growth rate, the global, annual FPV market will reach 6 GW of annual installed capacity and €9 billion in 2030. Significantly higher estimates also exist, such as from a USA government report of $27.7 billion by the end of 2031[26]. The great variation in predicted market size and growth rates also indicate the sensitivity of the growth to cost predictions for FPV. Analysis shows that FPV currently is 12-20% more expensive than GPV[27,28], and the rate of cost reductions will significantly influence the market size.

Realizing the 5-10% LCOE reductions targeted in SuRE will make FPV more cost competitive and more likely to reach the high market growth scenarios. By 2030, the difference in deployment resulting from an annual growth rate of 20% versus an annual growth rate of 30% will be more than 10 GW.

The uncertainty of the predictions highlights the sensitivity of FPV deployment with respect to future costs and policies. Sensitivity analysis shows that installed FPV system cost is most sensitive to float costs and module efficiency, followed by local loads[28]. Depending on system size, the structural balance of system (SBoS) cost constitutes 25% –30% of the total CAPEX. ***SuRE targets all the most critical parameters to reduce LCOE.*** The main technology developments in SuRE are directed towards reducing SBoS costs. In addition, we address optimization of module cost versus efficiency for FPV applications, and more accurate predictions of loads to reduce overengineering. For the business analysis carried out for this proposal, the consortium has predicted an annual growth rate of 25% as a benchmark scenario. The partners have analyzed how they expect the developments in SuRE to impact their growth beyond that of only following the market, as shown in Section 2.1.2. under "creating more and better jobs". We have not attempted to evaluate how SuRE can impact the growth of the total market for SuRE, although given the capacities, cost reduction and sustainability goals of the partners, it is highly likely that SuRE will not only increase the market shares of the partners, but also increase the deployment rate of FPV and expand the global market. Through competitive advantages related to LCOE, sustainability and synergies in exploitation, which will be realized through SuRE, all partners target a growth that is significantly greater than the benchmark market growth of 25%.

SuRE will enable the further deployment of FPV in Europe will be implemented with a substantially higher level of circularity and a substantial lower risk for the environment. Despite the potential, European countries have been slow to embrace FPV. The dominating market has been Asia. This is partly due to a lack of empirical data, but mostly due to the specific conditions that must be met – in terms of design, materials, system components, operation, and environmental impact – when working with bodies of water[29]. Long-term environmental sustainability is a vital requirement that FPV plants must fulfill if they are going to be approved for use in many European markets. The lack of knowledge on sustainability also infers that economic analysis lacks the full value proposition of FPV systems in detail because lifetime analysis of cost and revenues encompassing the impact of the energy-water-nexus is required to understand the complete economic feasibility of FPV applications[28]. SuRE is ambitious with respect to providing accurate and validated models for the impact of FPV on biodiversity and the local environment. The knowledge will be made available through accessible guidelines, directly helping the public authorities and regulatory bodies to make decisions based on high-quality data. Decreased costs, reduced uncertainties, and improved sustainability of the FPV systems, as achieved through SuRE are the factors needed for more rapid deployment in Europe. The potential is immense. In addition, the production capacity for both MetSolar, CTI, ZIM and Sunlit will be greatly enhanced, and CTI, MetSolar and Sunlit all plan new production lines to exploit the results in SuRE. The current module production capacity in Europe is limited, and additional production capacity is another important factor to ***enhance the security and autonomy of energy supply in Euorpe.***

2.2 Measures to maximize impact - Dissemination, exploitation and communication #@COM-DIS-VIS-CDV@#

SuRE will apply an impact-driven dissemination, exploitation and communication strategy consisting of three phases, designed to maximize the potential short-term outcomes and long-term impacts of the project and the wide scale roll-out of Key Exploitable Results (KERs). Compaz will be responsible for dissemination and communication, and a detailed Communication and Dissemination Plan (CDP) provided by M6 of the project (D7.1). The exploitation plan, provided by M12 (Exploitation Plan – D7.2) will have it's outspring from the Partners individual exploitation plans, but with joint consortium efforts to leverage the individual plans and collaborate for increased impact and uptake.

2.2.1 Dissemination strategy and measures

All partners, under coordination of IFE, will steer selection, elaboration, and creation of materials for dissemination. Compaz and IFE will collaborate closely with organizations that provide established and recognized channels for dissemination, including IEA PVPS, Solar Power

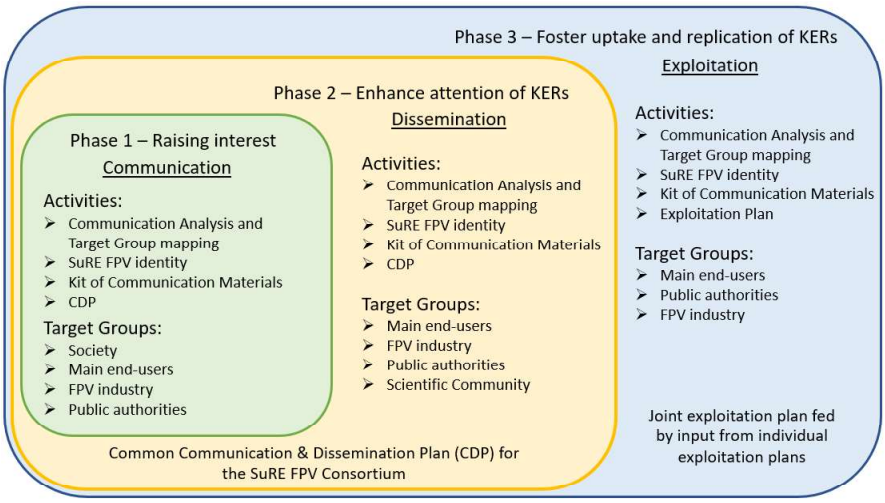


Figure 8: Overview of the Dissemination, Exploitation and Communication Strategy of SuRE

Europe and national clusters. The research partners will contribute to dedicated FPV workshops for industry and policy makers. Through collaboration with three universities (University of Oslo,University of Freiburg, Utrecht University) the research partners will engage in education and capacity building activites. Open access scientific publications, oral presentations and invited talks at conferences and industrial events will be used to share non-sensitive information to the FPV industry and research community.

The CDP will be developed that combine the Communication (Section 2.2.5) and Dissemination activies.

Dissemination Plan	
Activity	**Target groups**
Events (Workshops, Webinars, Conferences, Fairs)	
Host 2 events with international profile targeting main end-users. Objectives are: (1) to disseminate project results and project directions, (2) calibrate against decision frameworks of these actors.	Main end-users
Host 2 events with international profile targeting actors in the FPV industry. One is targeted to be co-organized with the Intersolar Exhibition in Munich. The other is targeted to be co-organized with the SolarPlaza FPV conference	FPV Industry
Host 4 workshops targeted at public (permitting) authorities. We will invite water authorities and other permitting authorirties such as provinces. Events will be organized in The Netherlands, Germany, France and Norway (where contacts and communication chanels are established).	Public authorities
Participation in 15 conferences and trade fairs. Objectives are: (1) interact actively with multiple stakeholders simultaneously, covering the European and international range. (2) The consortium showcases results achieved by the project through presentations, talks, exhibition spaces and personal engagement. Selected fairs and Industry conferences: Europe Solar Summit, Floating Solar Conference, Intersolar	Main end-users, FPV Industry, Public authorities
10 oral presentation at Scientific conferences. Selected scientific conferences we will attend: *EUPVSEC, Sophia reliability workshop, NREL reliability workshop*.	Scientific community
Publications (Peer-reviewed articles and conference proceedings)	
9 peer-reviewed articles in addition to conference proceedings. Selected journals for publication are: Solar Energy, IEEE Journal of Photovoltaics, Renewable Energy, EUPVSEC proceedings. Planned papers: 1) Modelling of FPV-water interaction and biodiversity, 2) Monitoring of water quality, 3) LCA of FPV Systems, 4) Field stressors for FPV, 5) Accelerated stress testing for FPV, 6) Operating temperature of FPV technologies, 7) Review and summary of WIML for tabulated sea states, 8) Review of soiling losses for FPV, 9) PLR for Utility Scale FPV power plants.	Scientific community, End-users, FPV Industry
1 White Paper expressly oriented to policymakers on the deployment of FPV. The publication outlines best practices, suggesting lessons learnt and measures to overcome obstacles. This document is designed to be easy-to-digest, short and straight to policy recommendations.	Public authorities
20 Awareness Publications. The project releases and contributes to over 20 publications in articles, blog posts, position papers, factsheets, and other non-scientific publications to reach a critical mass, targeting a less specialised public. Reference media: CORDIS Research*EU Magazine, AI-on-demand Web Cafés.	Main end-users, FPV Industry, Public authorities, Scientfic Community, Public.
Open Access Research	
SuRE facilitates an online repository to manage and publish material with different access permissions, including peer-reviewed publications and shareable scientific research data. The project relies on Zenodo, part of the European Open Science Cloud (EOSC), to deposit both publications and datasets. The use of Open Research Europe is encouraged as open access publishing platform for scientific articles among the consortium, without author fees and open access-compliance. Whenever possible, the datasets are made openly available via the data.europa.eu, the official portal for EU data.	Scientific community, Main end-users, FPV Industry, Public

2.2.2 Exploitation Strategy and measures

SuRE will make a strong effort and thus has dedicated activities to develop a business framework and market analysis leading to the exploitation and further implementation. All partners will be directly involved in these activities

bringing relevant options for market uptake. The role each partner plays in this strategy is different according to its nature. Exploitation activities will be carried out primarily in the individual companies, supervised by the project coordinator. WP7 will create effective measures for preparing the market uptake of the results, while the individual WPs deals with the development of the solutions themselves. The solutions will reach TRL6-7, thus, European regulations and directives will be fulfilled for the pilots, shortening market uptake. The SuRE consortium has defined an overall exploitation strategy describing how we will work to ensure that usable knowledge to be generated during the project, based on the three main steps below.

- Results Identification: Identification and definition of the key exploitable results (KER) of the project. This step also includes the identification of the level of interest in the different results and the identification of the added value of the project to boost further scientific and exploitation developments.
- Individual Partners' Exploitation Strategy: Definition of the partners' exploitation plans, including the market potential, the added value, the target end-users, the potential competitors, the expected impact on the partners' portfolio, a tentative timetable and the opportunities and barriers for the exploitation action.
- Consortium level Exploitation Strategy: Definition of common exploitation plan, based on a combination of the Individual Partners' Plans according to clustering criteria, including similar or common exploitation actions declared by the partners in their individual plans; the application domains considered for the exploitation; a consideration on the benefit expected from each exploitation actions.

Carrying out the exploitation strategy will lead to a concrete a periodically updated exploitation plan. This will facilitate the identification of new business opportunities and associated business models throughout the project. To identify KERs and select which results to pursue, feasibility studies to analyze the technical, financial, commercial and organizational feasibility of the proposed product will be useful. In SuRE, the main innovations and identified KERs are related to development of FPV technology. In the following, a preliminary feasibility study and business plan focused on FPV floater technology is provided.

2.2.2.1 Feasibility Study

Technical Feasibility. In Table 1.1.2, Section 1.1.2, 1.13 and 1.2 the technical feasibility of the planned innovations, the challenges and methods used to overcome them are thoroughly described.

Financial Feasibility. For the purpose and duration of SuRE, all partners have the capital needed to fulfil their obligations and leverage as tabulated in the budget. Sunlit will need investors to realize the planned ramp up of the production line in Norway. For this purpose, the Innovation Fund is a very relevant sources for funding. The next deadline for small investments in the Innovation Fund is fall 2023 fits well with Sunlit's first upscaling. For the technology developments planned in SuRE for CTI and Sunlit SuRE, funding for up-scaling will be required at a later stage (~2025-2026). Sunlit has significant experience with capital-raising and already has a network of subcontractors with capacity to deliver equipment for up-scaling. For the duration of the SuRE project the CTI and LKY have the necessary capital to realize the plans, including technology developments, the first production and pilots. For up-scaling, the production of the new technology, it will be necessary with capital raising. The Innovation Fund is one of several potential sources for this funding. It may also be that subcontracting and renting will be the best option, depending on available interest rates and conditions. BayWa, ZIM, SABIC, Metsolar and Compaz, have the necessary capital and infrastructure to realize the plans in SURE, including technology developments and pilot. For larger scale implementation and production of modules designed for FPV applications, MetSolar may require capital to invest in new tools and/or upscaling. Needs for further investments will depend on the results from SuRE.

Market Feasibility. The market potential for cost-competitive, and sustainable FPV technologies is immense and growing. We need to keep increasing the deployment rate of PV for many years to come if we are to meet average EU policy goals[1]. And as PV implementation expands FPV provides a vast potential of increased deployment without conflicting with other uses of land. Published estimates of the annual CAGR varies from 5% to 43%[25]. FPV is also predicted to be the segment of PV Market to grow at the highest CAGR from 2020 to 2025. A more thorough analysis of the market and market shares of the partners is provided in Section 2.1.4.

Organizational Feasibility. The partners and value chains are extremely well positioned to manage the extensive, foreseen growth. The technical knowledge, business insight, and value chains of CTI and ZIM already are world leading. Concrete plans for upscaling will be revised in the beginning of the project period. Sunlit has is currently working to scale up a very promising, patented technology and do their first scale up and investments in new production line with capacity 11 MW/year by early 2024. The founders of the company have many years of experience from the FPV business and from other start-ups prior to founding Sunlit. They will also benefit significantly from the deep knowledge of the FPV industry in the SuRE consortium.

2.2.2.2 Preliminary Business Plan

Business models are the main instrument to support the exploitation of SuRE's results. A business model creates value for companies by identifying how to convert their products into profit. At the proposal level the Consortium has decided to make use of the CANVAS model as a key strategic management tool allowing the description of a preliminary Business Plan for the three FPV technology providers and value chains. This will be further used, individualized, and improved as SuRE progresses. The Business Model Canvas (BMC) explains a business model graphically, by dividing it into 9 basic modules that reflect the logic that a company follows to raise revenue, covering the 4 main business areas: customers, value proposition offered, infrastructure and economic viability – finance. A first CANVAS model analysis has been performed for SuRE and is shown in Fig. 9.

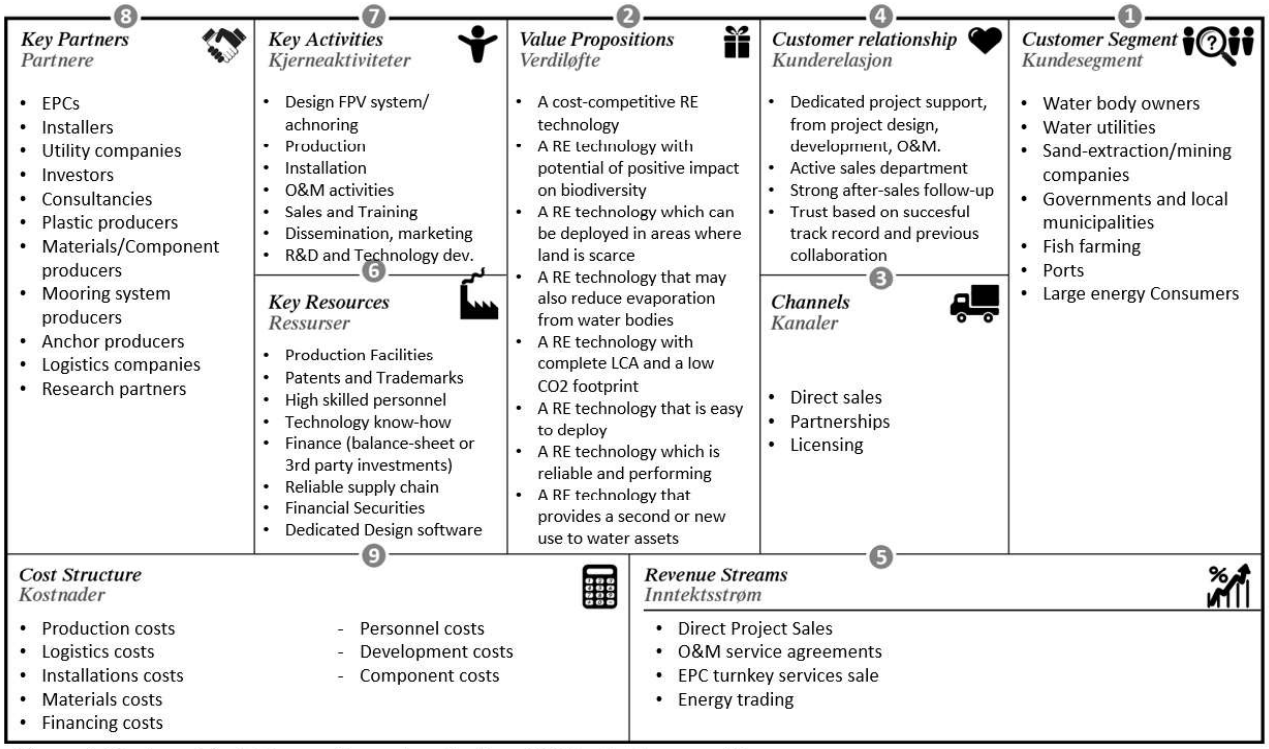


Figure 9. Business Model Canvas focused on the three FPV technology providers.

2.2.2.3 Exploitation Plan

Preliminary Individual partners' exploitation plan

Detailed individual exploitation plans for all partners will be developed within the first six months of the project period. A preliminary individual exploitation plan with the identified KERs is provided in the table below.

Table 2.5 Partner KERs and preliminary exploitation plans

CTI KERs: *New FPV technology and new product for efficient FPV maintenance.*
CTI will develop a new FPV product to be sold to utility companies and PV developers globally. The product is currently at TRL 3 and will be developed to TRL 6 by the end of the project (5 MW pilot will be TRL 5). Development will be done with the first off-taker, LKY, the four research institutes and a few central partners and subcontractors, one of which is SABIC. Quantified measurements from the 5 MW pilot will be important for the exploitation plan. Reports prepared by in SuRE on Sustainability, Reliability and Efficiency of the plant will serve as important documentation for new customers. **Target groups:** Utility companies/ PV developers, Policy makers, General public. **Main sales channels:** CTI sales department.
ZIM KER: *Modified FPV product for new applications.*
ZIM will expand the application area for the FPV technology from SSC 3 to 4. Quantified measurements from the pilot will be important means of documenting the enhanced structural integrity to potential customers. **Target groups:** 1) Main end-users, 2) FPV Industry, 3) Policy makers. **Main sales channels:** existing sales department
BayWa KERs: *Improved business case and extended range of applications for FPV projects.*
A reduced LCOE, higher reliability and higher sustainability of ZIM Float and other potential FPV technologies will make FPV more competitive and more appealing. This will be actively used in conversations with waterbody

owners, drinking-water utilities, hydropower operators and public authorities to increase their awareness and to boost the likelihood of success of BayWa´s sales activity, and hence the number and size of projects that can be deployed. In addition to BayWa's active participation in dissemination measures in WP7, BayWa will make use of direct conference calls, in-person meetings, workshops, and discussions with municipalities and local communities, industry conferences and invitation of stakeholders to projects and pilot sites. **Target groups:** 1) Main end users; Utility companies, Energy off-takers, water body owners, Investors. 3) Public authorities. **Main sales channels**: existing sales channels and relationships with investors, off-takers, water body owners. Tenders.

Sunlit Sea KER: *Improved integration of floater and module for FPV application*

Sunlit has developed an FPV based on an aluminum float with a solar panel laminated on top, serially connected in a string of floats. In SuRE we improve the product by reducing aluminum usage, strengthening the float to be more durable and open for use of recycled aluminum, assessing how the float handles rough sea and wind, improving the heat transfer to the water, and piloting proposed changes with sensors in a wave tank. The project will enable reduction in LCOE of up to 60% and scaling production from ~11 MW/year in 2024 to ~ 455 MW/year by the project end. It will enable certification of the design criteria for the product and thus open new markets for Sunlit. **Target groups:** 1) Main end users; Utility companies/PV developers, Investors 3) Public authorities 5) General public. **Main sales channels:** direct dialogue with main investors and potential customers.

Metsolar KER: *Design of PV module tailored for FPV.*

Metsolar is a custom PV module manufacturer and sees FPV as an application with great potential. The work in SuRE will be essential to quantify this potential by evaluating cost-efficiency of potential mitigation measures for FPV-specific stressors. MetSolar currently manufactures modules for Sunlit. Sunlit's scale-up plans would make them one of MetSolar's largest customers by project end. Cost optimizations and mitigation measures for FPV modules in general, and Sunlit's FPV technology, will be focus areas in MetSolar's exploitation plan. Changes to MetSolar's production line to allow more cost-efficient modules with larger cell-size is a likely optimization step to reduce cost of Sunlit's FPV technology that can also be exploited in MetSolar's other markets. **Target groups:** 1) Main end-users; Sunlit, PV industry. **Main sales channels:** direct dialogue with main investors and customers.

Sabic KER: *New low CO_2 processes and products for plastic FPV components.*

Sabic sees that the FPV market has strong growth potential. Sabic wants to achieve via SuRE a strong position to offer materials solutions for more sustainable FPV (improved circularity, lower CFP). The exploitation strategy of Sabic consists of (1) direct marketing towards FPV system providers, (2) influencing permit criteria by disseminating options on improved circularity towards permitting authorities. **Target groups:** 2) FPV industry. **Main Sales channels**: direct dialogue and development with potential customers.

IFE, TNO, Fraunhofer KERs*: Predictive energy yield models, accelerated stress tests, sustainability guidelines*

SuRE establishes the cooperation of the three leading European institutes in the field of FPV. For IFE, TNO and Fraunhofer, SuRE strengthens the partnership with the leading FPV companies in Europe and confirms the strong collaboration with SMEs in the field. IFE, TNO and Fraunhofer will gain valuable scientific and technical know-how that will allow the institutes to get an even stronger profile in the FPV field. The research institutes will exploit the project results through: i) Being the R&D partner for FPV industries; ii) Being the knowledge hub for public bodies; iii) Dissemination of results, create follow-up projects for next step innovations, also towards offshore FPV.

Deltares KERs: *Sustainability Guidelines, Software*

SuRE strengthen Deltares already leading position as knowledge hub and partner within sustainable deployment of FPV. Deltares also have a clear exploitation strategy for their software development, with the 'dare to share' principle. Deltares distribute Delft3D as 1) open-source code and 2) as a compiled version via a service package.

Joint exploitation plan

The joint exploitation plan will be completed, as a part of the overall exploitation plan, within the first 6 months of SuRE. SuRE is an ambitious and solutions-oriented project that can bring to market more affordable and sustainable FPV technologies. The SuRE consortium will make use of the Horizon Results Platform, a system developed by the European Commission to help promote the exploitation of the results from the R&I framework programmes. Using this platform will allow the partners to publish and promote uptake of their results towards the key target groups of SuRE. The key outcomes of will also enter the Horizon Results Booster (HRB), an initiative of the European Commission whose aim is bringing a continual stream of innovation to the market whilst maximizing the impact of public funded research within the EU. Through the HRB, SuRE will secure a larger Impact through an extended Dissemination and Exploitation, heading the research activities towards strong societal impact whilst concretizing the value of R&I activity for societal challenges.

2.2.2.4 IPR Strategy

Because SuRE comprises separate value chains of ostensible competitors, but whose common aim is to grow European FPV activities, it is critical to identify and isolate innovations which need to remain within one entity. Therefore, ownership and IPR management will allow each partner to innovate while protecting their IPR as set in the Consortium Agreement. Nonetheless, partners have already achieved a preliminary agreement on the IP strategy where each partner owns its IP background. The consortium has identified three levels of IP which will be created from the project: *(i) Individual and joint IP*, which belongs to individual partners or is jointly owned by partners working in a particular task and is restricted to those partners; *(ii) Generic IP*, which can be used by all partners of the consortium; *(iii) Publicly available IP* which will be published on the SuRE website and made available with no restrictions. Provisions for use of IP background will be determined during the commercialization strategy. Insights that enhance partners' services or enable the launch of new solutions within partners' strategic go-to-market plan will be kept private and protected via non-registered Soft IP measures (confidential information etc.).

2.2.3 Communication Strategy and Measures

The communication strategy of SuRE involves specific measures to promote the project itself and the results attained, with the mission to reach a broad audience. We will promote outreach and networking activities by collaborating with other relevant networks and projects to create joint dissemination activities thus further increasing the project's outreach. The communication and dissemination analysis also includes engaging and collaborating with our listed Target Groups as detailed in Section 2.1.1, as input to the concrete measures in the CDP. Common information and dissemination, sharing upon invitation by CINEA, to used increase the visibility and synergies between Horizon Europe supported actions. The CDP will be fine-tuned and closely monitored by the advisory board on a yearly basis.

Table 2.5 Method and channel of communication activities

Method	***Establish Communication Platforms***	***KPI target***
Communication and dissemination analysis	SuRE will analyse internal and external barriers to communicating and disseminating FPV progress. A detailed Target Group mapping will be carried out.	*CD analysis shared in the consortium*
Communiction and dissemination plan (CDP)	The communication action plan will be developed at the beginning of SuRE (M6), with yearly fine-tuning and close monitoring by the advisory board.	*CDP released*
Logo & Visual identity	The SuRE identity (logo, graphical chart) will be defined and a Kit of Communication Materials (website, brochure, template, flyers, video) will be developed to facilitate the CDP for both virtual and physical engagements of the consortium.	*Communication Kit-M6, Website & Social media accounts established in M3*
Method	***Activities Throughout the Project***	***KPI***
Website	The SuRE website will be the main communication tool to present our activities and consortium as a whole. A dedicated private space for partners will be implemented. All notable and public progress of SuRE will be published.	News – 1 Item monthly >10 000 unique views
Social and professional networking	Link SuRE with the professional network of consortium partners to share and discuss results, questions and get feedback. LinkedIn, Instagram and Twitter willl be used when relevant to raise interest in our activities.	>5000 followers total
Newsletters	The consortium will publish one e-newsletter each year.	1 yearly
Press media	The project coordinator will make press releases validated by SuRE partners. Partners are encouraged to contact media	3 yearly
Leaflets and Posters	We will produce visual material to promote SuRE and its activity at fairs, conferences, exhibitions, and other meetings.	4 leaflet versions; >1000 reach
Project movie documentary	Compáz will make a movie of around 20 minutes. It will tell the story of the FPV pilot development.	5000 views within 1 year of release

2.3 Summary

KEY ELEMENT OF THE IMPACT SECTION

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
➢ *The cost of electricity from FPV power plants must be reduced to accelerate the deployment of RE.* ➢ *There is potential to reduce the carbon footprint of the current FPV technologies.* ➢ *Uncertainties in predicted energy yield and reliability of FPV power plants is a hinderance to increased investments.* ➢ *Efficient regulation and permitting of FPV is hindered by the lack of quantifiable knowledge on the effect of FPV power plants on the local environment and biodiversity.* ➢ *Modelling tools to accurately predict LCOE from PV power plants are inadequate.* ➢ *There is not enough skilled labor to handle the increased installation rates required to reach the policy goals for PV.* ➢ *There is a lack of standards to access the reliability of components for FPV power plants.*	***New products & services:*** ✓ *Novel technology from CTI bringing LCOE down by 10%, including new maintenance strategy, is realized and verified in 5 MW pilot. Circularity is improved, and CO_2 footprint reduced.* ✓ *Reduced BoS of ZIM Float with 15% and extended application is realized and verified in 5 MW pilot. Reduced steel consumption and reduced CO_2 footprint with 20%.* ✓ *Sunlit floater with 60% LCOE reduction is realized and verified in 105 kW pilot.* ✓ *The potential of FPV technologies to positively impact the biodiversity is verified and quantified.* ✓ *The potential reduction in cleaning costs for FPV power plants is quantified.* ***New procedures:*** ✓ *Pre-normative standards for accelerated aging of FPV components.* ✓ *Concrete guidelines on how to evaluate the impact of an FPV power plant on the local biodiversity.* ✓ *Improved accuracy in EYA, applicable for FPV technologies (inland and offshore), implementable in commercial bankable tools.*	***Dissemination measures towards the 5 Target Groups:*** ➢ *1) Main end-users and 2) FPV industry; Participation to relevant industry commercial fairs and tradeshows, leveraging as well on meaningful relationship and increase networking activities, including the European PV Industry Alliance.* ➢ *3) Public authorities; dialogue meetings and white papers* ➢ *4) Scientific community; Publications with the results of the planned activities and tests; Participation at relevant conferences.* ➢ *5) General Public; See communication measures below.* ***Exploitation of Results from SuRE:*** ➢ *The three FPV value chains, involving the technical departments, and the market and sales departments each of the companies, will carry out the most critical work with respect to exploitation.* ➢ *All companies have efficient, existing channels for sales and marketing the new and/or improved products.* ➢ *Cross-cutting efforts across the value chains and individual partners will be vital to exploit the results related to sustainability, reliability and pre-normative standards and modelling tools.* ➢ *An innovation and Commercialization Board (ICB), with one employee from each partner will be central to maximize the exploitation of results towards public authorities and organizations, the scientific community and general public.* ***Communication towards a wide audience and the general public:*** ➢ *Social Media Profiles, visual identity (log), videos and a dedicated website. Organization of a final event open to a wider public to show the outcomes of the project, especially for the FPV industries, the scientific community, and policymakers.*

TARGET GROUPS	OUTCOMES	IMPACTS
➢ ***Target Group 1) Main end-users;*** *primary off takers of the technology developments and know-how created in SuRE. Utility companies, Engineering, Procurement and Construction companies and consultancy firms working within the field of FPV. Banks, funds, and investors in FPV – to reduce hesitancy of investing in FPV and reduce interest rates which greatly affects LCOE.* ➢ ***Target Group 2) FPV Industry.*** *Materials and components providers to the FPV industry will be a target group for work related to specifying requirements of different materials for various FPV applications.* ➢ ***Target Group 3) Public authorities.*** *Policy makers, concession granting bodies, standardization bodies, and other public authorities – particularly relevant off takers for the results on sustainability, recycling, and pre-normative tests for components.* ➢ ***Target Group 4) The Scientific community*** *will be a target group for the general knowledge on sustainability, recycling, predictive energy yield and reliability. Outreach to a broad scientific community is also important for training of new personnel.* ➢ ***Target Group 5) The public*** *will be target-groups for work on sustainability and public acceptance.*	***New FPV designs with reduced CAPEX and OPEX, greater energy output and reduced LCOE:*** ✓ *CTI is ramping up production of new FPV tech. and expanding the sales of new technology beyond LKY.* ✓ *ZIM Float has increased expanded their market and increased market shares outside of Europe.* ✓ *Sunlit are ramping up production in Norway.* ✓ *Optimized cleaning routines are picked up by the broader FPV industry and transferrable to other O&M providers/utilities.* ✓ *Models and performance evaluations of different FPV technologies verified and published.* ✓ *The perceived risks and uncertainties related to FPV is decreased. Investors and project developers have a better understanding of risk mitigation, electricity yield forecast and sustainability impact.* ✓ *Project developers and investors have an improved business case through the availability of improved FPV designs and a reduced LCOE.* ***The environmental impact of FPV technology for inland and offshore waters is verified to be low, or positive, for the local environment. FPV application is expanded:*** ✓ *Low-cost FPV technologies developed for low SSC expands their application to higher SSC.* ✓ *A good initial knowledge base for the effect of FPV power plants on [water quality, biodiversity etc.] is established and widely accessible.* ✓ *Monitoring procedures and methods to evaluate the environmental impact of an FPV power plant are successfully implemented for new FPV power plants.* ✓ *Permitting authorities achieve a better understanding of requirements/conditions for suitable waterbodies.*	***Economic:*** ✓ *Reduced risk and increased competitiveness of FPV power plants, inland and offshore.* ✓ *Deployment of FPV in Europe accelerate as investors and project developers have a better understanding of risk mitigation, predictive energy yield, improved business case through reduced LCOE.* ✓ *An increased market share for European FPV technologies/industry (CTI from 15% to 20%, ZIM from 7-12% and Sunlit from NA – 7%).* ✓ *A broader European value chain within FPV; Recyclable plastic components and cost-efficient PV modules for FPV applications produced in Europe.* ***Scientific:*** ✓ *New breakthrough scientific understanding of how FPV impacts local biodiversity.* ✓ *Guidelines adapted in the documentation requirements of concession bodies in three European countries.* ✓ *Increased competence of scientific and technical staff.* ***Societal:*** ✓ *Appropriate regulatory frameworks for FPV implemented in EU.* ✓ *Increased deployment rate of FPV in Europe, implemented with higher level of circularity and a substantially lower risk for the environment.* ✓ *Deeper integration of value chain partners/customers ensuring SuRE operation of FPV power plants.* ✓ *Dedicated accelerated aging tests for FPV components introduced in new IEC standard.* ✓ *FPV power plants widely acknowledged as sustainable and environmentally friendly RE technology.*

#§IMP-ACT-IA§#

3 Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

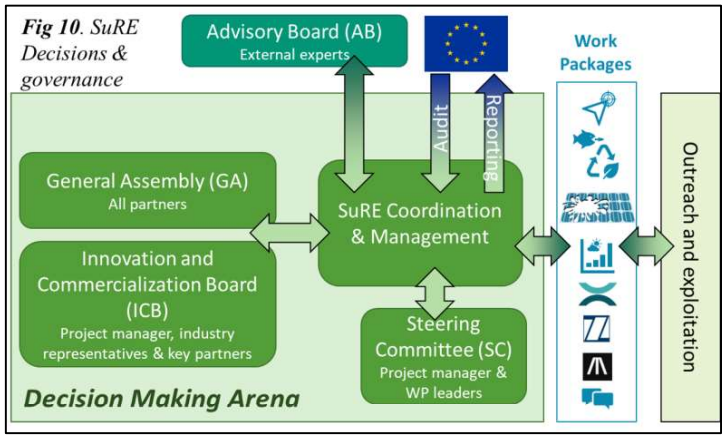


***Fig 10**. SuRE Decisions & governance*

The SuRE work structure promotes innovation development for all FPV segments while promoting targeted development of three commercial technologies. A members-only section of the SuRE website serves as a common platform for formal and informal communication. Formal project resources, such as contracts, planning, meeting minutes, and templates will be posted there. Project workflow and quality control will be shared internally as openly as possible, as closed as necessary, to foster transparent project management. Leaders of WPs 1-3 & 7-8 have recent experience of similar leadership responsibilities in EU projects. Leaders of WPs 4-6 lead internal and public R&D projects, and will receive guidance from the coordinator, internal and external assistants, and via the collaboration baked in to SuRE. All WPs run the full duration of SuRE.

Work distribution among WPs allows for decentralized activities and responsibilities while holding clear reporting and monitoring channels to align project management with achieving SuRE targets. Its structure is organized to reach the objectives by fostering FPV growth overall and applying that mindset to 3 commercial technologies. Figure 10 shows the SuRE management structure. Three bodies will promote smooth management while providing communication spaces for partners to express and discuss concerns and exchange project information: General Assembly (GA), Steering Committee (SC), and Innovation at Commercialization Board (ICB). In addition, 3 external experts, from ***Dutch national water authority***, ***DNV***, and ***Total Energies***, respectively, have agreed to serve as an Advisory Board (AB) to SuRE. The advisory board will be an organ to discuss fine-tuning of and uptake of results in the broader FPV industry, for review of material, and a source of knowledge and connections with other ongoing initiatives within FPV.

General Assembly (GA). The GA facilitates WP coordination and opportunities to discuss major changes to the working plan. One representative from each partner will sit on the GA, meeting 4 times a year. It is the decision-making body for major SuRE issues like changes of coordinator, work plan, budget, partners, IPR considerations, and other general non-technical matters.

Steering Committee (SC). The SC is the operational management body and is responsible for efficient implementation. The SuRE coordinator, WP leaders, and the commercial members of the ICB constitute the SC. The SC works under the GA and assists, makes proposals for, and reports to, the GA. It monitors and ensures technical quality progress, approves progress reports, coordinates review of deliverables, assesses milestones, and moderates technical issues that span multiple WPs. The SC monitors ethics in SuRE.

Innovation and Commercialization Board (ICB). The ICB supports knowledge transfer to end-users, elevating exploitation and commercialization opportunities that allow the IPR owners to fulfil SuRE Outcomes. Knowledge transfer plans will be developed, according to type, application, IP conditions, and end users. ICB participation is mandatory for commercial partners, and optional for others; maximum 1 participant per partner. The coordinator is the ICB chairwoman.

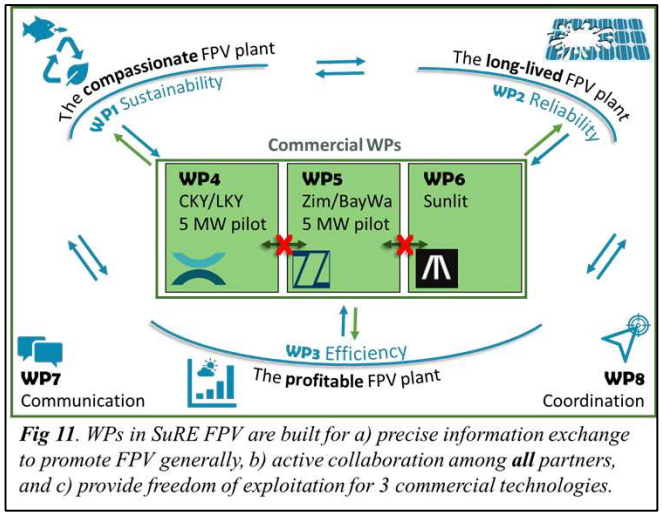


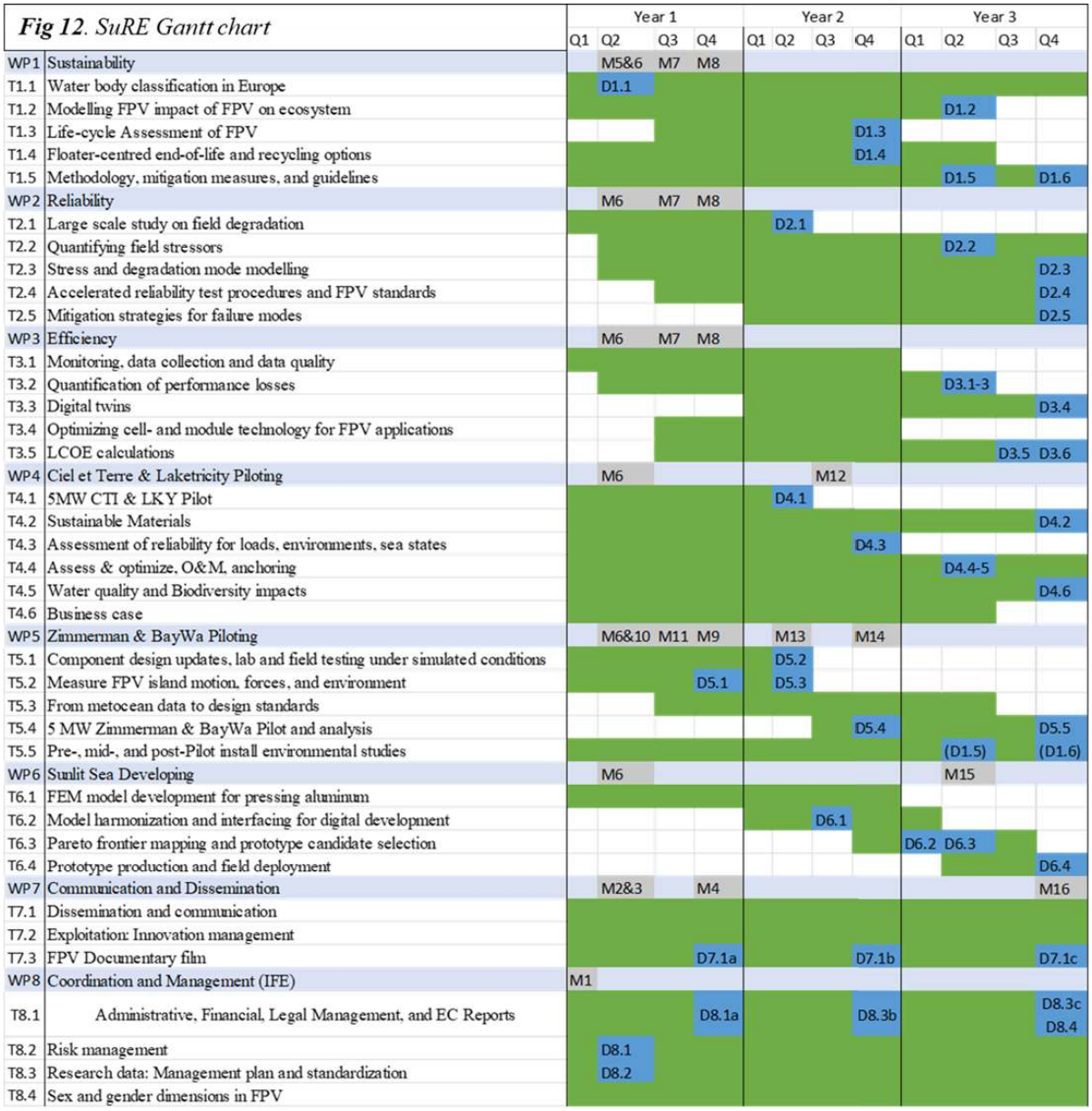
***Fig 11**. WPs in SuRE FPV are built for a) precise information exchange to promote FPV generally, b) active collaboration among **all** partners, and c) provide freedom of exploitation for 3 commercial technologies.*

3.1 Work plan and resources

As shown in Figure 9, WPs 1-3 push FPV boundaries in Sustainability (*compassionate* FPV), Reliability (*long-lived* FPV), and Efficiency (*profitable* FPV). The work here is *generalizable* across FPV, pushing boundaries for the whole field, yet also includes participation of the commercial partners. In the interior, WPs 4-6 apply the generalizable methods to the 3 specific FPV technologies of CTI, ZIM and Sunlit. The work here is *specific*, yet also includes participation of the research institutes. Maximal information exchange along the circumference of WPs 1-3 and radially into WPs 4-6 is how we can reduce CAPEX, OPEX, and LCOE while increasing environmental benefit. Minimal information exchange among the

central WPs 4-6 provides independence to each technology developer to exploit their own results.

Surrounding and supporting these WPs are WP8 to coordinate and manage, and WP7 to communicate and disseminate. WP8 also takes the role of controlling and facilitating information flow. The coordinator, who is well-versed in commercialization communication, information protection, and multi-party R&D, will structure and be responsible for SuRE info handling. WP7 gains access to all other WPs to extract information for sharing; WP7 is also subject to information-flow control of the coordinator (WP8) and the owners of the information.



#§CON-SOR-CS§# #§PRJ-MGT-PM§#

Table 3.1a: List of work packages

#	Title	Lead #	Lead	PMs	Start	End
1	Sustainable FPV Power plants	9	Fraunhofer	107,8	1	36
2	Reliable FPV Power plants	8	TNO	101	1	36
3	Efficient FPV Power plants	1	IFE	94	1	36
4	Development and piloting of new CTI FPV tech.	2	CTI	179,0	1	36
5	Development and piloting of ZIM FPV tech.	4	ZIM	80	1	36
6	Development of Sunlit's integrated FPV tech.	6	Sunlit	59	1	36
7	Dissemination & Communication	11	Compaz	36,5	1	36
8	Coordination & Management	1	IFE	48	1	36

Table 3.1b: Work package description

WP number	**1**
WP title	**Sustainable FPV Power Plants**

Objectives

O1.1) Determine the interaction of FPV with ecosystem biodiversity
O1.2) Develop methodologies and guidelines to assess the effects of FPV on aquatic systems and biodiversity
O1.3) Consider potential mitigation measures to counter negative impacts of FPV
O1.4) Methods development for impact quantification by enhancing existing tools and monitoring strategies
O1.5) Publish guidelines for sustainable FPV systems, including mitigating environmental impact.

Description of work

Task 1.1: Water body classification in Europe (M1-M36)
Lead: Deltares. Participants: Deltares, IFE, LKY, BayWa, ZIM, Sunlit
Commercial partners and Deltares will combine existing catalogue of FPV-amenable European waterbodies. Then, select waters using European WFD classification and describe those suitable for FPV according to waterbody types (e.g., shallow lakes, deep reservoirs, intertidal zones), protected areas based on WFD and N2000, the existing ecosystem services (e.g., drinking water, recreation, nature, water supply). The result is a decision support tool for waterbody selection used in Task 1.2 & 1.3 and initial input to guidelines in Task 1.8. Fraunhofer will develop a monitoring concept for FPV-water interaction. This includes monitoring hydrological variables such as oxygen levels and water temperatures, air temperatures, and wind speed. Deltares develops a monitoring concept for FPV-aquatic food web interaction. Both monitoring concepts are finalized in consultation with IFE, ZIM, LKY, and Sunlit. Commercial partners then integrate sensors based on finalized monitoring concepts into FPV demonstrators in WP 4-5. Results are the monitoring protocol, early data are for calibration and validation of modelling FPV-water interaction in Task 1.3, and 3-year dataset for Tasks 1.3-1.5.

Task 1.2: Modelling FPV impact on ecosystem (M1-M30). Lead Fraunhofer. Participants: Deltares, IFE
We determine maximum FPV coverage (using ratio of blocked irradiation, mRBI) for a wide range of typical European waterbodies. At less than mRBI, environmental impact conforms to Water Framework Directive 2000/60/EC water quality and biodiversity criteria. In addition, waterbody evaporation reduction will be computed as a function of RBI. First, Fraunhofer, Deltares and IFE will calibrate their models with existing data from the demonstrator sites or comparable sites elsewhere. Fraunhofer will run exploratory simulations in their GLM tool to determine approximate mRBIs and evaporation reductions. Deltares will run detailed Delft3D simulations guided by GLM-derived mRBIs to compute more refined values and with several FPV designs. Together with IFE, evaporation reductions are computed, and results are compared with GLM 1D analysis to assess accuracy gains. SuRE partners calibrate and validate their results with monitoring data from WPs 4-6. Results are enhanced, validated tools to assess the direct impacts of FPV on aquatic systems (GLM and Delft3D), catalogue of maximum permissible FPV coverage ratios (mRBIs) differentiated by waterbody type and several standard FPV designs, and dependence of evaporation reduction on FPV coverage ratio for typical European waterbodies.

Task 1.3 Life-cycle Assessment of FPV (M6-M36)
Lead: TNO. Participants: Fraunhofer, TNO, Deltares, SABIC
TNO will provide a base model for LCA that quantifies CO2 footprints for current FPV designs using procedures EF 3.0 and/or ReCiPe. Fraunhofer and Deltares will propose additional LCA metrics that capture FPV impact on waterbody and biodiversity. Fraunhofer and TNO carry out LCA analyses for a variety of standard FPV designs, including those used in demonstrators in WP 4-6. The LCA analysis of FPV systems will be enhanced by a cradle-to-grave approach and additional impact categories such as biodiversity. SABIC contributes its expertise in plastic components of FPV installations. TNO and Fraunhofer link to IEA Photovoltaic Power System Program Task 12 and project(s) funded by HORIZON-CL5-2022-D3-03-09 for latest PV recycling concepts. Results are enhanced LCA methodology for FPV, assessment of ecological impact of updated FPV design throughout their life cycle.

Task 1.4 Floater-centered end-of-life and recycling options (M1-M30). Lead: SABIC. Participants: TNO
SABIC will lead the establishment of the most cost-effective, sustainable end-of-life, and recycling options for plastic components in FPV designs used in SuRE. The goal is to improve recyclability of plastic floaters and other components used in WP 4-6. Fraunhofer and TNO will use existing libraries for remaining FPV components, including PV modules, cables, connectors and inverters. Results are financially feasible and sustainable end-of-life strategies for major FPV designs.

Task 1.5 Methodology, mitigation measures and guidelines to assess environmental impact of FPV(M1-36)
Lead: Deltares, Participants: All

Deltares conducts scenario studies of high-yield, low-impact FPV, preferably for demonstrator sites in WP 4-6 and option to include similar sites elsewhere. Studies include sessions with stakeholders, technicians, landscape designers, and ecologists to evaluate perception of different designs in the context in the three pillars of sustainability: Residents, direct users of the water body (e.g., recreation), and public at large, governments, policy makers; Producers and energy companies; Environmental targets (WFD N2000) and nature conservationists. Results are definitions of barriers and enablers for FPV with a positive impact, and pathways to implementing nature inclusive FPV designs. Based on results Tasks 1.1-1.5, Deltares and Fraunhofer will propose adjustments to FPV to increase sustainability. Numerical simulations of a) methods to increase light penetration without compromise on PV yield, b) plant location evaluation within waterbody to foster fish stock, and c) added structures specifically for aquatic life. Consequences of proposed design changes for durability, LCOE, and electric yields will be numerically evaluated in WPs 2 & 3 and refined with commercial partners. Results are a list of options for project demonstrators to enhance sustainability and for generalizability in pre-normative standards. Finally, Fraunhofer and Deltares will compile guidelines for development and implementation of FPV solutions with a minimally negative and/or maximally positive environmental impact. The guidelines incorporate and summarize the learnings and results of all WPs. Results are published guidelines for design and implementation of sustainable FPV solutions.

WP number	**2**
WP title	**Reliable FPV Power Plants**

Objectives

O2.1) Perform systematic large-scale study on field failures and stress profiles from portfolio of FPV assets
O2.2) Develop simulation models to support understanding of degradation root causes and help system design
O2.3) Develop and publish recommended accelerated reliability test procedures for components, and use these to evaluate mitigation strategies for component failures

Description of work

Task 2.1: Large scale study on field degradation (M1-18)

Lead: TNO. Participants: IFE, Fraunhofer, LKY, CTI, BayWa, ZIM, Sunlit

Statistics on degradation modes will be collected from within and outside the consortium. The consortium has access to a broad collection of system designs, deployment conditions, and associated degradation modes for both inland and near shore water bodies. Information on degradation modes will be collected from field observations and data analyses, aerial thermography, combined with time-series analysis for degradation mode detection and power loss quantification (when possible), visual inspection, and field IV curve tracing, and site-sporadic bird soiling. Results are key degradation modes and statistics, used as input for definition of accelerated lifetime test procedures and mitigation of degradation modes for Tasks 2.3-5.

Task 2.2 Quantifying field stressors (M3-36). Lead: IFE. Participants: All

Gathering field data on FPV-specific stress profiles by instrumenting sites with at least two different floater technologies with specific sensors to assess mechanical loads (using strain gauges and accelerometers), salt exposure (for the near-shore conditions), humidity, and component time of wetness. Focus areas are mechanical loads on hinges, cables, inverters, and PV modules; salt exposure of modules, junction boxes and connectors; local humidity levels at inverters and underneath modules for pontoon-based structures; and time-of-wetness at junction boxes, module edges and connectors. Results are FPV-specific stress profiles for each component to validate stress and degradation mode modelling in Task 2.3, and to guide accelerated stress test development in Task 2.4

Task 2.3 Stress and degradation mode modelling (M3-36). Lead: IFE. Participants: Fraunhofer, CTI, ZIM, Sunlit

Develop multiscale modelling chain for assessing mechanical load and wear on individual FPV components in relevant design scenarios. We will carry out model developments for, and establish proper interfaces between, three state-of-the-art modelling tools: (1) Aero-hydro-servo-elastic simulation software 3DFloat (IFE), made originally for floating wind turbines using the FPV-relevant structural and hydrodynamic modules. (2) Computational fluid dynamics (CFD) modelling software Ansys Fluent (IFE) / COMSOL (Fraunhofer) for wind loads. (3) Finite-element modelling (FEM) software COMSOL (Fraunhofer) / SiSim (IFE) for detailed studies on PV modules.

For 3DFloat, several modelling adaptations and improvements will be assessed and implemented as necessary to accurately represent the FPV design scenarios. For structure, we will test replacing finite element beam sections with basic shell elements to represent floating components more accurately. For waves and wave-structure interactions, we will test the need for more advanced and tailored wave spectra, as well as n-body problem-based potential theory modelling to capture neighbor interaction and array effects.

FEM will be used to study humidity ingress into PV modules and other electrical components, based on humidity stress profiles gathered in Task 2.2. The results will be compared with results from accelerated stress testing in Task

2.4 and GPV modelling to assess the need for extended or adapted humidity stress testing. FEM-based thermal modelling (Fraunhofer) will assess hotspot formation from bird droppings, based on case studies selected from sites monitored in the project. Results will be calculation of mechanical loads/stresses, humidity ingress and hotspot susceptibility for the different designs scenarios. This will feed into the stress test development in Task 2.4, as well as into the technology-specific WPs 4-6. From the loads/stresses both component ultimate limit states and fatigue limit states will be established.

Task 2.4 Develop accelerated reliability test procedures and recommended FPV-specific standards (M7-M36)
Lead: Fraunhofer. Participants: All
Design and conduct meaningful laboratory tests to extend PV components testing to FPV challenges using degradation mode statistics from Task 2.1, stress profiles from Task 2.2, and simulation results from Task 2.3. Target components include PV modules, cables, connectors, and power electronics. We will focus on FPV-specific stressors: mechanical, moisture/salt stress, and hotspots caused by isolated soiling such as bird droppings. Preliminary submersion tests towards qualifying panels for FPV. Selected components will be subjected to test procedures of single stressors, combined, or sequential stress tests. Results are the published test procedures and defined test criteria., input for Task 2.5. The results from Tasks 2.1-2.5 will be summarized to create pre-normative test guidelines for FPV equipment. The result is published state-of-art requirements and tests for FPV components, designed to foster co-development with component suppliers thereby de-risking future FPV.

Task 2.5 Assess, design, and apply mitigation strategies for failure and degradation modes **(M13-M36)**
Lead: TNO. Participants: Fraunhofer, IFE, CTI, ZIM, BayWa, Sunlit, MetSolar
Improved mitigation strategies will be assessed for effectiveness in several steps. One, from degradation and stressor findings of Tasks 2.1-3, we will determine one or more mitigation methods for modules and components and/or their combinations. Two, by using accelerated ageing results from Task 2.4, test for failures by applying the accelerated testing to components with and without mitigation strategies. Purpose-made samples will be manufactured by TNO. Results are recommended improvements for material, components, and their combinations.

WP number	**3**
WP title	**Efficient FPV Power Plants**

Objectives
O3.1) Increase bankability for new FPV systems through more accurate EYAs
O3.2) Set up a simulation framework that enable identification of potential design improvements
O3.3) Enable improved O&M of FPV power plants by combining the models developed into a digital twin.

Description of work
Task 3.1: Monitoring, data collection and data quality (M1-M24)
Lead: IFE. **Participants:** TNO, Fraunhofer, CTI, LKY, BayWa, ZIM, Sunlit
The aim of this task is to ensure that the data-basis for carrying out the work inn all the WPs in the project is as good as it possibly can be. Production and monitoring data from real sites is indispensable to validate models and extract reference values. Monitoring- and production data from power plants, pilots, and test sites of different sizes, will be collected from all partners, including production and monitoring data from 18 sites owned by CTI/LKY, 3 sites owned by BayWa, and 3 sites owned by Sunlit. Data from these sites will be quality controlled, and additional monitoring equipment will be installed if needed. The research partners also have ownership of data from various smaller FPV systems that will also be utilized in the project. This includes a well instrumented Ciel & Terre system in South Africa, TNOs test site in the Netherlands and Fraunhofers's German test site.

Task 3.2: Quantification of performance losses for FPV (M3-M30)
Lead: IFE. Participants: TNO, Fraunhofer, CTI, LKY, BayWa, ZIM, Sunlit, Metsolar
To assess thermal loss, CFD models previously developed at IFE and Fraunhofer will a) be developed to include evaporative cooling, and b) be using the specific geometries and material properties of the FPV technologies as input. Together with production and monitoring data (Task 3.1) the models will be used to analyse efficiency of the different cooling mechanisms (convective, radiative, conductive, evaporative), as a function of climatic parameters. Recommended practice on how to include thermal losses of FPV power plants in the most widely used commercial tool, PVsyst, will be published. To assess PLR and soiling loss rates, the work will follow two routes previously developed for ground-based PV and published by IFE[19] and Fraunhofer. One is based on an analytical, iterative approach ("Combined Degradation and Soiling loss, CODS"), the other is based on machine learning on order to decompose performance index time series into a soiling component, a degradation component, and a seasonal component. Further model development will be to adapt filtering routines to account for FPV specific losses, such as WIML or local soiling variation or other FPV specific signatures. To assess WIML, TNO and IFE will model mismatch and compared to monitoring values for the technologies in the consortium at selected sites. Fixed POA is

translated to moving POA irradiance given different sea states and the movement of the respective FPV technology. The hydrodynamic models will be developed or adapted in WP2. From this, an electrical model will be used to calculate the mismatch in each string, and the effect of different inverters/maximum power point trackers is analysed. Finally, a wave-induce mismatch corrected Performance Ratio is calculated.

Task 3.3: Digital twins (M12-M36)
Lead: IFE. Participants: Fraunhofer, TNO, CTI, LKY, BayWa, ZIM, Sunlit, Metsolar
With the major loss mechanisms for the three FPV technologies identified, simulated, and quantified through work in Tasks 3.1 and 3.2, the work in this task will combine them to make a digital twin to study interdependencies and potential improvements in the design, and potentially to continuously benchmark the live performance of the power plants. The work will be done with two levels of complexity. First, a digital twin of the PV power plant will be built in python. This is based on modifications of existing code developed at IFE and Fraunhofer. The parameterization of thermal losses (U-values) and WIML that has been developed in Task 3.2 will be used to reduce the processing time and interaction between different types of models. Representative soiling loss rates and PLRs based on results from Task 3.2 will also be used as input. Second, we will explore a more complex digital twin combining the actual simulations performed in Task 3.2, this will be done for one FPV technology. This digital twin is expected to be computationally heavy, and we will study and quantify the inaccuracies inferred by various simplifications applied to the individual simulations.

Task 3.4. Optimizing cell- and module technology for FPV applications (M6-M24)
Lead: MetSolar. Participants: IFE, TNO, Sunlit, BayWa, CTI, LKY
Production costs of different cell and module technologies versus the resulting efficiency in the field will be analysed. Field efficiency will be calculated for the three FPV technologies in the consortium and applications in Europe. Then, we will evaluate the cell- and module selections that contribute to the lowest LCOEs. Next, MetSolar will consider adaptions in their production-lines to accommodate the optimization. The choice of adaptions must follow from the analysis, but three likely pathways have been identified; 1) prismatic glass to adapt to low incident angles for FPV, 2) change in cell size to M10/M12 to have higher efficiency modules to offset the greater BoS cost, and 3) adaptions to reduce PRL and improve reliability for FPV applications, based on results from Task 2.5.

Task 3.5: LCOE calculations (M6-M36)
Lead: TNO. Participants: Fraunhofer, CTI, LKY, BayWa, ZIM, Sunlit, Metsolar, IFE.
This task will set up the methodology for LCOE calculations to be used as a foundation when proposing options for cost-effective engineering solutions throughout the project and to enable comparative LCOE estimations. LCOE is strongly dependent on location, type of water body and choice of technology, so a bottom-up approach[28] will be used. To consider the impact on LCOE for different use cases, a set of parameters such as sea state, locations, water depth, etc. will be defined. For each parameter, the relation with the different components of LCOE calculations is defined. Correlation between given parameters and energy yield and degradation is found in Task 3.2, whilst the respective impact on CAPEX and OPEX is done together with the FPV technology providers within the consortium.

WP number	**4**
WP title	**Development and piloting of new CTI FPV technology**

Objectives
O4.1) Test and verify basic design of new CTI technology
O4.2) Reduce LCA further through assessing recycled materials with lower carbon footprint
O4.3) Optimize design for next generation of the technology

Description of work
Task 4.1: CTI Pilot of* at least *5 MW (M1 to M18). Lead: CTI. Participants: LKY
First large-scale installation (5 MW pilot) of fundamentally new CTI FPV technology, with reduced LCOE and improved LCA compared to the existing CTI FPV technology, Hydrelio, and installation follow-up (the installation itself will be done by CTI's customer and subcontractors). A pre-pilot of 50kW will be first installed to assess the new system design and validate its deployment for the 5MW pilot. A standard SCADA and monitoring system will be installed by the customer while IFE, TNO, Fraunhofer, and Deltares will contribute with installation of additional sensors packages for biodiversity, performance, and reliability of structure and anchoring. Data from the pilot will be used to perform measurements and verify models in WP1, 2, 3, and 4. The pilot forms an essential basis for further developments, as described in Tasks 4.2-8.

Task 4.2 Sustainable materials (M1 to M36) Lead: CTI. Participants: SABIC, TNO
Using ongoing learnings from pilot (Task 4.1), integrate and validate sustainable materials. Start with selecting materials meeting criteria of sourcing, mechanical fitness, water-stable, toxicity, and sustainability. Testing

performed for ageing, mechanical strength, and compatibility with industrial production. Materials used in pilot (Task 4.1) will be compared to more sustainable (such as recycled) materials in order to improve performance and carbon footprint. CTI will collaborate with SABIC (WP1) and TNO for ageing tests (WP2).

Task 4.3 Assessment of reliability for different loads, environments, and sea states (M1 to M24)
Lead: CTI. Participants: TNO, IFE
Measurements of mechanical loads and strain on structures and mooring/anchoring of the 5 MW pilot will be used to validate and adapt models developed prior to pilot installation. Use best-known accelerated stress tests from WP2 to perform CTI-specific ageing and fatigue tests. Complimentary mechanical tests (tensile tests, ageing, fatigue, and corrosion tests) will be performed with CTI internal lab, and in collaboration with TNO and IFE in Tasks 2.2 and 2.3. Results are quantified by showing a) reduction of over-design and b) subsequent LCOE reduction. A bench development will allow to qualify CTI pilot's structure in relation to its interactions with the panel.

Task 4.4 Assessing and optimizing the next generation of the CTI design, including O&M optimizations and Anchoring (M1 to M36). Lead: CTI. Participants: LKY, Fraunhofer, IFE, TNO, Deltares
Inputs from work on anchoring, O&M optimization, Tasks 4.1-4.3, WP2 (mechanical strength) and WP3 (losses, LCOE) will be used to assess and optimize the new system to reach TRL6. Anchoring simulations with CTI internal model for interaction between the floating structure and its anchoring system to understand mechanical loads passing through the structure and the anchoring system. Considerations include wind pattern for FPV design to include gust size applied for FPV anchoring design. In parallel, IFE will do the same with their own models. Comparisons among the models will allow model improvement and optimizing the anchoring system design used on the pilot (Task 4.1) and go from TRL 4 to 6/7. The design modifications are planned to be installed on next generation CTI product. For efficient O&M, CTI has developed a bespoke maintenance device which will be tested on the pilot. To date, an experimental proof of concept exists, and target is to improve it for use in the operational environment of the pilot (from TRL3 to 7). Finally, CTI, with input from SABIC, Fraunhofer, TNO, IFE, and MetSolar, will assess CAPEX, OPEX, LCOE and LCA of the 5 MW plant as a benchmark, and then co-optimize the individual considerations and improvements studied in Task 4.1-5 against each other. This includes everything from the ease of the industrial production processes of the chosen design to logistics, mechanical strength and expected lifetime, O&M, and expected production.

Task 4.5 Water quality and biodiversity impact (M1 to M36) Lead: CTI. Participants: LKY, Deltares, Fraunhofer
To assess water quality and toxicity, CTI/LKY will monitor changes in water at different water strata and assess its chemical composition. CTI, Fraunhofer, and Deltares will use sampling methodology developed in WP1, including pre-, mid-, and post-install of the pilot (Task 4.1). CTI and LKY will also contribute with data from existing FPV plants in Japan. To assess impact on biodiversity, Deltares will pre-assess the pilot using ecological inventories based acoustic recording, visit of experts and DNA samples. Inventory will cover the species on the site (avifauna, pisciculture, entomofauna, flora) to have an exhaustive analysis. A sizing of biodiversity enhancement measures will also be carried out, as examples, for installation of ponds for amphibians and walls for swallows. For monitoring, Deltares will visit the site pre-, mid- and post-install to measure the FPV plant and enhancement-measures impacts. Data sharing with WP1 to include in generalizable results, modelling, and guidelines.

Task 4.6 Business case. Lead: CTI. Participants: LKY (M1 – M30)
CTI and LKY will study the business environment related to the project outcomes. The task will start from the beginning of the project and will be closely aligned with research and pilot activities. Objectives will be a) to explore the marketing opportunities, b) to analyze the drivers and barriers for the adoption of the technology among stakeholders of the whole value chain, c) to define new business models, and d) share the achievements (project realizations) in the proper communication channels to show and expand the potential market. Business interest, market attractiveness and business models will be evaluated. The task will feed the Consortium exploitation plan (Task 7.2), while sensitive outcomes will remain internal to LKY and CTI.

WP number	**5**
WP title	**Development and piloting of Zimmermann FPV technology**

Objectives
O5.1) Concept development ZIM Float for rougher waters & environment to increase standard design principles to SSC4 (1.5-2 m significant wave hight)
O5.2) Reduce CAPEX and OPEX by adequate design and selection of BOS components and materials
O5.3) Build 5 MW pilot that combines WP1-3 best practices with ZIM Float and BayWa system design

Description of work
Task 5.1: Component updates, lab and field testing under simulated conditions (Hardware) (M1-M18)

Lead: ZIM. Participants: BayWa, TNO, IFE
Modify component designs for use on flexible island with SSCs 2-4. Inputs are conditions identified in Tasks 5.2-3. Use computational and prototype tests of new materials and/or concepts to test structural pieces that link island parts. Tests include force, vibration, corrosion, cyclic cable bending, inverter and transformer shaking and swinging, bending construction and connectors, material interaction, bio-interaction and biofouling, and system-mooring connectors. Testing conditions are humidity, salinity, cyclic loading based on results of Task 5.2. Results are top candidates to be tested /implemented in Tasks 5.3-4.

Task 5.2: Measure island motion, forces, and environment (M1-M18)
Lead: BayWa. Participants: ZIM, IFE, TNO
ZIM and BayWa will identify existing plants for installing wave, motion, force, and ambient sensors. Sensor type, deployment method, and analysis will be developed with IFE and TNO. Inputs are from WP2. Targets are quantifying environmental stressors, load cycles, and deriving how degradation factors interact. Candidate factors are global island movement, humidity, salinity, and sunlight; candidate failures are in inverters, transformer stations, modules, and connectors. Results are analysis of data which will are input to Tasks 5.2-4.

Task 5.3: From met ocean data to design standards (M6-M30)
Lead: BayWa. Participants: ZIM, IFE, TNO
Use literature review and measurements from Task 5.2 to connect water body categories with environmental conditions and FPV design. Assemble optimized components in Task 5.1, measurements in Task 5.2, and WP1-3 results to make decision tree for faster, standardized FPV design. Incorporate optimized FPV island flexibility to reduce material use, decreases lifetime costs, and adhere to sustainability procedures. Results are the candidates to be tested in pilot Task 5.4 and information feedback/feedforward with Tasks 5.2-3 and WP1-3.

Task 5.4: ZIM/BayWa Pilot of* at least *5 MW and subsequent analysis (M18-M36)
Leads: BayWa. Participants: ZIM, IFE, Deltares, Fraunhofer, TNO
Build 5 MW pilot using up-to-date advances from SuRE. Incorporating best practices of own technology development from Tasks 5.1-3 and WPs 1-3. Pilot contains extra sensors for added research benefit and accelerated expansion to higher SSCs, choice of sensors defined in WP1-3. Results are customer-connected pilot with future access for measurement and research. The first annual cycle will be used to assess improvements and verify models and methodology developed in SuRE.

Task 5.5: Pre-, mid-, and post-Pilot install environmental studies (M1-M36)
Lead: Deltares. Participants: BayWa, ZIM, Fraunhofer
Use pilot site to implement environmental impact tests on a specific FPV project to show general applicability. Use methodologies of Task 1.6 and early draft guidelines of Task 1.8 to establish protocols and measurements at pilot site before, during, and after installation of the pilot from Task 5.4. Results are timeline study, recommended improvements to guidelines of Task 1.8, and recommended improvements to future FPV.

WP number	**6**
WP title	**Development of Sunlit's integrated FPV tech**

Objectives
O6.1.1) Developing a digital product development cycle for Sunlit's aluminum floater
O6.2.1) Optimized selection of new floater generation prototype

Description of work
Task 6.1: FEM model development for pressing of aluminum (M1-M24). Lead: IFE. Participants: Sunlit
Model development specifically for WP6 will be on aluminum-float pressing, central to Sunlit manufacturing. IFE will extend its existing FEM modelling ecosystem to handle numerically challenging large metal deformations that occur in this process. The new tool, FormSim, will be developed as a module for existing tools SiSim and AlSim, where additional algorithms will be implemented to handle large mesh deformations and for anisotropic yield stress model. Further, code developments will implement higher order plasticity elements, robust triggers, thresholds for remeshing, and error bounded remeshing. Results are used in Tasks 6.2 (and further in 6.3-4)

Task 6.2 Model harmonization and interfacing for holistic digital product development (M12 – M26)
Lead: IFE. Participants: Sunlit, TNO
Digital product development cycle implementation, using all key design parameters of the aluminum float. Sunlit guide establishing and implementing interfaces between the following modelling tools: i) Combined FEM and CFD model chain to get mechanical stress profiles on the float for a given design case, developed in WT2.3 (IFE). ii) CFD modelling of fluid and heat flows, developed in WT 3.2 (IFE). iii) FEM model of the aluminum pressing, developed in WT6.1 (IFE). iv) LCA model developed in WT1.3 (TNO). Especially important is harmonizing input

and output data from the different models to enable automated mapping of the parameter space and rapid iteration turnover. Results are modelling capabilities at different accuracy-levels will be used actively to quickly screen the parameter space and converge to promising design candidates. Results are used in Task 6.3 (and further in 6.4)

Task 6.3 Pareto frontier mapping & prototype candidate selection (M20 – M32)
Lead: Sunlit. Participants: IFE, Fraunhofer, Metsolar
Use modelling framework from Task 6.2 to efficiently map Pareto frontier of float-design parameter space. Key design parameters for exploration because of co-impact on sustainability, reliability, and cost are (1) Reduced aluminum thickness and/or aluminum feedstock quality, such as with recycled aluminum; 2) Modified floater shape for increased mechanical robustness in rougher waters and/or for lower aluminum consumption; (3) Changed floater size to reduce unit costs (from cabling, factory operations, PV cell and module, etc.) per square meter; (4) Improved cup design to achieve more homogeneous water cooling for improved performance and reliability. This work will happen in concert with Metsolar, who will harness the modularity of the PV industry and match module to float design. Further, Sunlit's bill of materials compendium will help identify more- and less-expensive alternatives to today's production. New, market-available optical components, PV cells and prismatic glass, will be examined to further push these fronts. Results are prototype designs from Pareto analysis, used in Task 6.4.

Task 6.4 Prototype production and field deployment (M28 – M36) Lead: Sunlit. Participants: IFE, Metsolar
Informed by models and design process developed and carried out in Tasks 6.1-6.3, prototypes will be built. Protype testing in subcontracted wave-tank with extra sensors will validate simulated results. Sunlit will manufacture small press tools using existing machinery built to test the most critical parts of stretch forming. The best design will be the basis for the next generation press tool. The press tool will be built and installed in a large hydraulic press to make full-size product. Financing for the tool is outside of SuRE, the Innovation Fund will be considered. Final wave-tank tests will verify the design. Sunlit will build an (approximately) 100 kW pilot installation by the end SuRE (Singløya Nord, Norway) with guarantees of future access to production data SuRE research partners.

WP number	**7**
WP title	**Dissemination, Exploitation and Communication**

Objectives
O7.1) Targeted strategic communication that strengthen and expand synergies
O7.2) Timely and appropriate dissemination of SuRE outcomes to appropriate Target Groups
O7.3) Coordinate innovation efforts and use consortium strengths to maximize exploitation of KERs

Description of work
Task 7.1 Develop and implement communication and dissemination plan (CDP) (M1-M36)
Lead: Compáz. Participants: All
The task will be focused on the definition, coordination, and implementation of the CDP. In the first six months, the SuRE identity (logo, graphical chart) and the Kit of Communication Materials (website, brochure, template, flyers, video) will be developed. The first CDP, D7.1, will be released at M6, with subsequent updated plans in M18 and M30. Social network accounts for the project will be set up. From M6 and onwards, the framework developed, will be filled with content and regularly updated to inform about project news. The partners will attend the targeted conferences, publications, lectures, outreach events and workshops as detailed in Sec 2.2.1. Compaz will perform Communication activities as detailed in Section 2.2.3 and track the Dissemination activities. All partners have planned activities to contribute to communication and dissemination of SuRE results. In addition to Compaz and IFE, BayWa, with their strong network among water-body owners, drinking water utilities, hydropower operators and public authorities, will have a particularly active role within the dissemination work.

Task 7.2 Exploitation and Innovation management (M1-M36). Lead: IFE. Participants: All
IFE implements innovation management to allow SuRE to respond to external and internal opportunities to exploit creative efforts to introduce new ideas, services, processes, and products. Key elements have been listed in Sec 2.2.3. Key SuRE innovation management is monitoring innovation actions and supporting partners to enhance their own innovation capacity to integrate new knowledge. Task 7.2 identifies companies' go-to-market needs and to provide support to enhance their own potential for impact delivery. Innovation management will be closely related to the project tasks on exploitation and defining the key exploitable results. The consortium partners will have individual exploitation plans feeding into the Consortium exploitation plan delivered in M12. Innovation management will be discussed in every General Assembly, with concessions for limited-audience conversation when required.

Task 7.3 FPV SuRE short films (M1-M36). Lead: Compáz. Participants: All
Compáz will create yearly 2-min short films about the project goals and the global objectives and vision of the project. The films will be produced and released from the start of the project each year. The content of these videos

will be the communication line and content of the communication activities and diffused on the social media. It will serve SuRE in its communication activities to reach larger audiences beyond FPV professionals.

WP number	**8**
WP title	**Coordination and Management**

Objectives

O8.1) Coordinate SuRE; oversee R&I and ensure communication between partners
O8.2) Project management: Quality assurance, time planning, reporting (according to CA), and financial control
O8.3) Anticipation and evasion of risks, problems, and conflicts
O8.4) Definition, implementation, and compliance of data management plan
O8.5) Elevate novelties for fair allocation of IPR according to Consortium Agreement

Description of work

Task 8.1 Admin, Financial, Legal Management, and European Commission Reports (M1-M36). Lead: IFE
Project management is via consolidated management of the different processes (costs, resources, communication, content and delay, risks, purchase, legal, and excellent). PM includes managing interactions among partners, WPs, activities, external relationships, and managing modification and deviations. Task 8.1 includes management of overall SuRE progress through meetings and technical and financial reporting. Reports will be collected from WP leaders on 6-month intervals. WP8 will manage preparations and dissemination of reports. IFE, as coordinator, will ensure communication and reporting with EC, including any potential major issues or changes in the work plan. IFE will oversee establishment and monitoring of governing boards.

Task 8.2 Risk management (M1-M36). Lead: IFE
Risk management ensures proper risk handling and prediction, impacts estimation, and mitigation execution. Each participant, from partners to the General Assembly (GA), is responsible for continuous potential risk monitoring and reporting to the coordinator to allow early identification and mitigation. Identified risks will be brought to the GA for amelioration.

Task 8.3 Research data: Management plan and standardization (M1-M36). Lead: IFE
SuRE is composed of a broad group of researchers, commercial actors, and communication specialists. We will produce large and heterogenous datasets, some of which will be proprietary, others obfuscated to protect its origins, and others for the public domain. Further, we will also generate several statistical and mechanistic models and the results of their implementation. Task 8.3 will develop and implement the SuRE data management plan according to FAIR principles; current recommendations or standards provide guidelines for big and small datasets, codes and models. IFE will generate and distribute metadata templates supporting FAIR handling and completion using the public resource Norwegian Centre for Research Data (NSD). Common procedures and standard methodologies will follow international guidelines for maximal value and comparability. The Management Board (MB) will agree and circulate written guidelines and standard operating procedures for implementation. Task 8.3 provides continuous FAIR documentation and data handling support to all WPs.

Task 8.4 Sex and gender dimensions in FPV (M1-M36) Lead: IFE. Participants: All
Kickoff, mid-point, and closure meetings include a workshop session with external advisor who will address innovation and workplace aspects leading to improved gender equality at all levels of PV industry.

Table 3.1c: List of Deliverables

No	*Deliverable name*	*Short description*	*WP*	*Lead*	*Type*	*Dissemination level*	*Delivery date (month)*
1.1	Monitoring Concepts	Protocol for measuring hydrological and ambient variables, including waterbody selection	1	Fraunhofer	OTHER	PU	M6
1.2	FPV coverage	Publication on FPV coverage and impact on environment	1	Deltares	R	PU	M30
1.3	LCA report	LCA (CO_2/materials/biodiv.) CTI/ZIM	1	TNO	R	PU	M24
1.4	Plastics recycling	Recommended process and material for plastic components in FPV	1	SABIC	R	PU	M24
1.5	Biodiversity mitigation	Review of FPV designs for biodiversity and potential pathways to nature inclusive FPV designs.	1	Deltares	R	SEN	M30
1.6	Guidelines FPV	Final FPV sustainability report w/	1	Fraunhofer	R	PU	M36

	Sustainability	potential for biodiversity increase.					
2.1	Failure modes	Statistical analysis of failure modes	2	TNO	R	SEN	M18
2.2	Field stressors	Quantification of field stressors	2	IFE	R	SEN	M30
2.3	Degradation	stress and degradation mode modelling	2	IFE	R	PU	M36
2.4	Accel. stress	Accelerated reliability test for FPV	2	TNO	R	PU	M36
2.5	Mitigation strategies	Report on assessing and applying mitigation strategies for failure modes	2	TNO	R	PU	M36
3.1	WIML Report	Report on annual mismatch losses	3	TNO	R	PU	M30
3.2	Thermal model	Paper on FPV Thermal Modelling	3	IFE	R	PU	M30
3.3	Soiling and PLR	Report on PLR and soiling loss rate	3	IFE	R	SEN	M30
3.4	Digital twin	An operating digital twin	3	Fraunhofer	DEM	PU	M36
3.5	CTI LCOE	Report on LCOE of CTI	3	Fraunhofer	R	SEN	M32
3.6	CTI ZIM	Report on LCOE of Zimmermann	3	Fraunhofer	R	SEN	M36
4.1	CTI's pilot (v.1)	5MW pilot installed w/ monitoring	4	CTI	DEM	SEN	18
4.2	Report sustainable mat.	The report will gather the tests results and industrial results	4	CTI	R	SEN	36
4.3	Report on tech. Validation	The report will gather the tests results and industrial results	4	CTI	R	SEN	24
4.4	Specific O&M device	A system prototype demonstration in operational environment	4	CTI	DEM	SEN	30
4.5	Optimise anchoring sim.	Report on software model with optimized parameters	4	CTI	R	SEN	30
4.6	Environmental study report	Report gathering results of the analysis of biodiv. and water quality	4	LKY	R	SEN	36
5.1	Load model	Accurate Hydrodynamic model of loads on connectors & steel structure	5	ZIM	R	SEN	12
5.2	Design spec. pilot	Specifications of materials, connectors, anchoring solutions for varying SCCs	5	ZIM	R	SEN	18
5.3	Corrosion testing	Report on steel coatings for salt-water applications	5	ZIM	R	SEN	18
5.4	BayWa/ZIM pilot	5MW pilot installed w/monitoring	5	BayWa	DEM	SEN	22
5.5	Field stressors	Measurements at BayWa/ZIM pilot	5	BayWa	R	PU	26
6.1	Sunlit model chain	Data-format and inter-model data-transfer specifications	6	IFE	R	SEN	M20
6.2	Sunlit Pareto	Pareto Frontier mapping, identification of promising prototype designs	6	Sunlit	R	SEN	M24
6.3	Sunlit BOM	Module cost and efficiency improvements for Sunlit's design	6	Metsolar	R	SEN	M30
6.4	Sunlit Pilot	~ 100 kW pilot installed w/monitoring	6	Sunlit	DEM	PU	M34
7.1	SuRE Film	3 x 2 min film from SuRE and FPV	7	Compáz	DEC	PU	12,24,36
8.1	Risk & mana-gement plan	Document describing management of SuRE and its risk	8	IFE	R	PU	M6
8.2	Data man. plan	Data management plan of SuRE	8	IFE	DMP	PU	M6
8.3	Progress Report (v.1, 2, 3)	Management, Technical, Innovation and Gender Progress Reporting	8	IFE	R	PU	M12,24, M36
8.4	Final report	Final report detailing SuRE results	8	IFE	R	PU	M36

Table 3.1d: List of milestones

Milestone number	Milestone name	WPs	Due date M	Means of verification
M1	Kick-off meeting	8	3	Physical Meeting
M2	CDP released	7	6	Plan published online.
M3	Communication Platforms	7	6	Website with logos/graphical profile online
M4	Exploitation Plan	7	12	Joint exploitation plan circulated internally
M5	Link established to FP in CL5-	1	6	Meeting with coordinator, plan for joint

	2022-D3-03-09			workshop
M6	Access to existing data CTI/ZIM/Sunlit sites	1 - 6	4	Data collected and shared with RP
M7	Sites for extra monitoring equipment defined	1-3	8	3 existing PPs selected for installation of additional monitoring equipment
M8	Install additional sensors	1 -3	12	Sensors installed on 3 existing power plants
M9	Scope of sustainability studies defined	5	12	Mutual understanding what studies, with what scope and timeline will be performed
M10	Met-ocean conditions	5	6	Mapped environmental factors for SSC2,3, 4
M11	Critical Components identified	5	10	System fully screened for components that needs to be re-design to SSC 4 locations
M12	Data from CTI 5 MW pilot	4	20	Pilot up and running
M13	ZIM/BayWa 5 MW pilot	5	18	Permits granted, pilot design fully defined
M14	Data from ZIM 5 MW pilot	5	22	Pilot w/monitoring equipment installed
M15	Data from Sunlit pilot	6	30	Pilot w/monitoring equipment installed
M16	E-Plan complete	7	34	Exploitation Plan Completed

Table 3.1e: Critical risks for implementation #@RSK-MGT-RM@#

Description of risk (i) likelihood, and (ii) severity: Low/Medium/High	WPs	Proposed risk-mitigation measures
Deviation of the schedule at critical points (low/high)	ALL	Stringent project management and good coordination in the work packages
Lack of data from sites to use in for model calibration and/or validation in WP1, 2 and 3. (low/medium)	1,2,3	Using literature values, use research sites. Install monitoring equipment at new sites.
Lack of data for comprehensive LCA (medium/medium)	1	Adapt impact categories to available data, obtain data from literature
Modelling results too coarse for efficient use (low/medium)	4, 5, 6	Reallocating resources to work more with model, or adapt solution with a safety margin
CTI's client reject the new O&M tech. as not mature enough (TRL) for the 5MW pilot. (medium/medium)	4	Use a less innovative but more mature and based on the same principles.
Recycled materials have negative impacts on water or are too expensive (medium/high)	4, 6	Change processing or material
Influencers or activities may take a stand against the project (low/high)	4, 5	To deploy more efforts on communication activities
Pilot site permits could be postponed or not granted => delay forecasted activities (medium/high)	4, 5	Other sites are available in both CTI/LKY and ZIM/BayWa portfolio
Material availability (low/high)	4, 5, 6	Second sourcing and early commissioning
Insufficient standards that are applicable (low/high)	4, 5, 6	Derivation of procedures of other norms or DNV recommended practice

Table 3.1f: Staff efforts #$RSK-MGT-RM$#

Staff efforts	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	Total PM/ Participant
1. IFE	2	37	29	9	6	15	6	28	132
2. CTI	2	13	2	121	0	0	2	1	141
3. LKY	2	1	17	28	0	0	1	1	50
4. ZIM	3	3	3	0	44	0	1	3	57
5. BayWa	3	3	3	0	23	0	4,5	1	37,5
6. Sunlit	1,5	2	3	0	0	27	1	1	35,5
7. MetSolar	1	4	6	1	0	11	1	1	25
8. TNO	10	24	18	8	4	4	1	3	72
9. Fraunhofer	29	12	12	3	3	2	2	4	67
10. Deltares	51,3	0	0	5	0	0	1	1	58,3
11. Compáz	0	0	0	0	0	0	15	3	18
12. SABIC	3	2	1	4	0	0	1	1	12
Total PM	**107,8**	**101**	**94**	**179**	**80**	**59**	**36,5**	**48**	**705,3**

Table 3.1g: Subcontracting costs

Subcontracting costs	Cost (€)	Description of tasks and justification
CTI	9000	Installation of monitoring equipment and sensors on 5 MW pilot
CTI	60000	Processing of material for next generation recycled plastics
CTI	75000	Static and cyclic mechanical testing of 5 MW pilot BoS
CTI	30000	Production of the new O&M device
CTI	62000	Simulation of anchoring for pilot site, benchmark simulations for further optimization of anchoring solutions in the consortium
CTI	30000	Static and cyclic mech. testing of updated technology (after 5 MW pilot)
LKY	60000	Water quality measurements for existing sites
ZIM	35000	Weathering and tensile and sharpy test of different materials
ZIM	60000	Static and structural design and verification, fatigue simulation and FEM
ZIM	25000	Corrosion testing and verification
BayWa	52000	Derive met-ocean conditions and adaption of the anchoring to SCC4
BayWa	45000	Testing of cables and inverters for vibration and movement
Deltares	38400	Education and guidance of the PhD by Utrecht University

Table 3.1h: Purchase costs (OGWS: Other goods, works and services)

Participant: 2/CTI	Cost (€)	Justification
Travel and subsistence	19 267	WP8: Project meetings (2x5 meetings), WP2, WP4: Technical meetings WP7: Participation in dissemination activities and international fairs.
Equipment	230 000	50kW R&D pilot at first, amortized over 3 years (100000€), O&M device procurement (50 000€), optimization trials (300 000€), optimized anchoring systems procurement (20 000€), prototype parts (BoS, cable and electrical equipment supports) (30 000€)
OGWS	35 000	Patent fees and CFS
Participant: 4/ZIM		
Travel and subsistence	15 000	Travel to project meeting and sites
Equipment	25 000	Cost for sensors and monitoring equipment including installation
OGWS	28 000	Material costs, testing of connectors and mooring connectors. CFS.
Participant: 5/BayWa		
Travel and subsistence	12 000	Travel to project meetings, fairs/dissemination activities, pilot site
Equipment	75 000	Sensors for acceleration, vibration, humidity, and evaporation
OGWS	5 000	CFS
Participant: 6/Sunlit		
Travel and subsistence	4 500	1 person, 4 meetings abroad (1 meeting in Norway)
Equipment	15 000	Monitoring equipment, pilot
OGWS	157 000	Materials (aluminum, steel) software license, CFF and logistics
Participant: 7/MetSolar		
Travel and subsistence	5 000	1 person, 4 travels
Equipment	55 000	Tools to handles M10/12 wafers, depreciated over SuRE lifetime.
Participant: 8/TNO		
Travel and subsistence	25 000	Project meetings, field visits, conference
Equipment	67 000	Laboratory equipment, TNO field lab
OGWS	45 980	Consumables for module manufacturing and components, CFS

For the partners IFE, LKY, Deltares, Fraunhofer, and SABIC, purchase costs are under 15% of the personnel cost. No in-kind contributions from third parties and other costs categories are envisioned for SuRE.

3.2 Capacity of participants and consortium as a whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

SuRE has 12 partners from 6 European countries, composed of 4 research institutes and 8 companies, as presented in Chapter 1. The project will build **two 5 MW FPV demonstrators** plus one smaller, yet commercially relevant, demonstrator. The pilot sites will be situated in Belgium (CTI/LKY, building and connection permits in place), Netherlands (BayWa) and Norway (Sunlit). As we describe in the introduction Sec. 1.1.1, **every member of SuRE works actively in FPV**. The research partners in SuRE are at the international forefront of FPV research and already collaborate actively with the FPV industry, including SuRE partners. This is also evident from the list of FPV related projects in Part A of this proposal. The institutes have recognized publications within the field of FPV and are

regularly invited to present their work at international conferences, workshops, and fairs. CTI is the second largest producer of FPV technology globally measured by installed capacity, ZIM/BayWa is the leading FPV technology provider/installer in Europe with a market share of 60%, and Sunlit have patented a promising, novel, FPV technology which they are rapidly scaling up. SABIC and Metsolar support FPV technology development by providing necessary expertise in the components. All partners are commercially active, R&D active, and committed to environmental positivity of FPV. Compáz, with experience at the nexus of communication-public engagement-solar energy, puts SuRE in the limelight to showcase the technical, social, and environmental focus of our work. We are prepared, and exceptionally well positioned, to solve the challenges posed in the call: elevate the environmental position of FPV, expand the application and decrease the LCOE. Every partner in SuRE has worked with at least one other partner on previous FPV activities, as shown in Part A. SuRE has been built up based on competence, dialogue and close cooperation between the partners, without mediators or consultants taking care of writing the proposal. The project starts from a good understanding of the partners' roles, tasks, and responsibilities and all of them have actively contributed to shaping the activities of SuRE.

IFE has experience as coordinator in H2020 and Horizon Europe, and has a dedicated European Affairs Department, who will assist on financial matters and reporting. IFE's Coordination Team allocated for this project will consist of 3 people, including principal scientist, Dr. Josefine Selj as technical coordinator, 1 European advisor (who will manage the financial and EU reporting aspects) and the Head of European Affairs and Business Development of IFE. This setup allows effective contact with the European Commission while enabling scientific progress. Organization of SuRE WPs, as described in Sec 3.1, is designed to allow all partners to complement one another. Major laboratory and production infrastructure are in place at the sites required; SuRE is a method to share expertise and resources without needing to double-up on equipment or training. Such collaborative exchanges are baked into the project design as given in the WP description in Table 3.1b.

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